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[Intervention Review]

Recombinant follicle-stimulating hormone (rFSH) versus other recombinant or urinary gonadotropins for ovarian stimulation in assisted reproductive technology cycles

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ABSTRACT

Rationale

Ovarian stimulation is an essential step during in vitro fertilisation (IVF) and intra-cytoplasmic sperm injection (ICSI), needed to produce multiple follicles for follicle aspiration. The strategy of stimulating the ovaries with gonadotropins is well established and can be performed using different types of gonadotropins. This review update brings together all randomised studies that compare recombinant follicle stimulating hormone (rFSH) with urinary gonadotropins and other recombinant rFSH (including rFSH from a human cell line) and biosimilars.

Objectives

To compare the effectiveness and safety of recombinant follicle-stimulating hormone (rFSH) with the three main types of urinary gonadotropins, rFSH derived from a human cell line (follitropin delta) and rFSH biosimilars for ovarian stimulation in women undergoing IVF or ICSI treatment cycles.

Search methods

We carried out an extensive search on 31 March 2025, according to Cochrane guidelines. The databases searched included: the Cochrane Gynaecology and Fertility Specialised Register of controlled trials, CENTRAL, MEDLINE and Embase. We also searched trials registries, reference lists and contacted experts in the field for additional studies.

Eligibility criteria

We included all randomised controlled trials reporting data comparing clinical outcomes of women undergoing IVF/ICSI cycles and using rFSH in comparison with human menopausal gonadotropin (HMG) or highly purified HMG (HP-HMG), purified urinary follicle-stimulating hormone (FSH-P), highly purified urinary FSH (FSH-HP), follitropin delta, and rFSH biosimilars for ovarian hyperstimulation. We used the Trustworthiness in RAndomised Controlled Trials (TRACT) checklist to assess trustworthiness and excluded potentially problematic studies from the analyses.

Outcomes

We implemented the core outcome set for infertility. Our main outcomes were live birth, ovarian hyperstimulation syndrome (OHSS) (both critical outcomes) and clinical pregnancy.

Risk of bias

We used the Cochrane risk of bias tool (RoB 1) to assess the risk of bias.

Synthesis methods

We combined data using a fixed-effect model to calculate an odds ratio (OR) or mean difference (MD). We summarised the overall certainty of evidence for the main outcomes using GRADE criteria.

Included studies

We included 59 studies (25 new studies) with a total of 18,119 women undergoing fertility treatment.

Synthesis of results

rFSH versus HMG/HP-HMG

Live birth is probably lower with rFSH compared to HMG/HP-HMG (OR 0.83, 95% CI 0.73 to 0.95; 15 studies, 4793 participants, moderate-certainty evidence).

OHSS is probably higher with rFSH compared to HMG/HP-HMG (OR 1.42, 95% CI 1.12 to 1.80; 37 studies, 9813 participants, moderate-certainty evidence).

Clinical pregnancy is probably lower with rFSH compared to HMG/HP-HMG (OR 0.87, 95% CI 0.76 to 0.98; 15 studies, 4839 participants, moderate-certainty evidence).

rFSH versus FSH-HP

Live birth: there is probably little or no difference in live birth between rFSH and FSH-HP (OR 1.02, 95% CI 0.86 to 1.21; 15 studies, 2956 participants, moderate-certainty evidence)

OHSS: there is probably little or no difference in OHSS between rFSH and FSH-HP (OR 1.01, 95% CI 0.67 to 1.53; 18 studies, 3640 participants, moderate-certainty evidence).

Clinical pregnancy: there is probably little or no difference in clinical pregnancy between rFSH and FSH-HP (OR 1.00, 95% CI 0.88 to 1.14; 22 studies, 4535 participants, moderate-certainty evidence).

rFSH versus follitropin delta

Live birth: there is probably little or no difference in live birth between rFSH and follitropin delta (OR 0.88, 95% CI 0.74 to 1.04; 3 studies, 2688 participants, moderate-certainty evidence).

OHSS is probably higher with rFSH compared to follitropin delta (OR 1.62, 95% CI 1.18 to 2.23; 4 studies, 2988 participants, moderate-certainty evidence).

Clinical pregnancy: there is no difference in clinical pregnancy between rFSH and follitropin delta (OR 0.94, 95% CI 0.80 to 1.11; 3 studies, 2688 participants, high-certainty evidence).

rFSH versus biosimilars

Live birth is probably higher with rFSH compared to biosimilars (OR 1.29, 95% CI 1.09 to 1.54; 8 studies, 2883 participants, moderate-certainty evidence).

OHSS: there may be little or no difference in OHSS between rFSH and biosimilars (OR 0.77, 95% CI 0.57 to 1.06; 8 studies, 2883 participants, low-certainty evidence).

Clinical pregnancy was probably higher with rFSH compared to biosimilars (OR 1.35, 95% CI 1.14 to 1.60; 7 studies, 2783 participants, moderate-certainty evidence).

The certainty of the evidence ranged from low to moderate for live birth and OHSS, and from moderate to high for clinical pregnancy. The main limitations were risk of bias in the included studies – unclear selection bias and selective reporting, as well as high risk of other bias. Many studies were sponsored by the pharmaceutical industry.

Authors' conclusions

Live birth and clinical pregnancies are probably lower with rFSH compared to HMG/HP-HMG, while OHSS is probably higher with rFSH when compared to HMG/HP-HMG. There is probably little to no difference in live birth, clinical pregnancy, or OHSS between rFSH and FSH-HP. When compared with follitropin delta, there is probably little or no difference in live birth or clinical pregnancy, while OHSS is probably higher with rFSH. When compared with biosimilars, live birth and clinical pregnancies are probably higher with rFSH, while there may be little or no difference in OHSS.

Funding

This Cochrane review had no dedicated funding.

Registration

Protocol (2005) <https://doi.org/10.1002/14651858.CD005354>

Review (2011) <https://doi.org/10.1002/14651858.CD005354.pub2>

PLAIN LANGUAGE SUMMARY

Which hormones are most effective for stimulating the ovaries in women undergoing in vitro fertilisation (IVF)?

Key messages

- Live births are probably slightly decreased with recombinant follicle-stimulating hormone (rFSH) compared to purified (HMG) or highly purified (HP-HMG) human menopausal gonadotropin, but probably increased with rFSH compared to biosimilars (manufactured gonadotropins). There is probably little to no difference in live births between rFSH and purified follicle-stimulating hormone (FSH-HP) or follitropin delta. Clinical pregnancy is probably decreased with rFSH compared to HMG/HP-HMG but probably increased with rFSH compared to biosimilars. There is probably little to no difference in clinical pregnancy with rFSH compared with FSH-HP and follitropin delta.
- OHSS is probably higher with rFSH compared to HMG/HP-HMG and follitropin delta, but there is probably little to no difference with rFSH compared to FSH-HP, and there may be little or no difference with rFSH compared to biosimilars.
- Future research should focus on frozen embryos and cumulative outcomes across multiple frozen embryo transfer cycles to give a more complete picture of treatment effectiveness. Strategies for the prevention and management of OHSS should be prioritised.

What is in vitro fertilisation?

In vitro fertilisation (IVF) is a type of fertility treatment. Ovarian stimulation helps women's ovaries produce more eggs, which are collected, fertilised and transferred back to the woman. Various hormones, including gonadotropins, are used to stimulate the ovaries:

- purified human menopausal gonadotropin (HMG) and highly purified (HP-HMG);
- purified follicle-stimulating hormone (FSH-P) or highly purified follicle-stimulating hormone (FSH-HP);
- recombinant follicle-stimulating hormone (rFSH) was developed in a laboratory to obtain higher purity of the hormone;
- follitropin delta was developed to work in a more personalised way; and
- 'biosimilars' are manufactured to offer similar results to rFSH at a potentially lower cost.

Ovarian hyperstimulation syndrome (OHSS) is an unwanted and sometimes serious effect of IVF, in which the ovaries swell and fluid can leak into the abdomen.

What did we want to find out?

We wanted to know which gonadotropin produced more live births, caused less OHSS, and led to more 'clinical' pregnancies, confirmed by ultrasound.

What did we do?

We searched for studies that compared one type of gonadotropin with another for women undergoing IVF. Studies could use fresh or frozen embryos.

What did we find?

We included 59 studies with a total of 18,119 women undergoing fertility treatment in a clinic.

rFSH versus HMG/HP-HMG (17 studies, 5639 women)

- Live births probably decrease with rFSH compared to HMG/HP-HMG. If the chance of live birth with HMG/HP-HMG is 28%, then the chance of live birth with rFSH is between 22% and 27%.
- OHSS is probably higher with rFSH compared to HMG/HP-HMG. If the risk of OHSS with HMG/HP-HMG is 2.6%, then the risk of OHSS with rFSH is between 2.9% and 4.7%.
- Clinical pregnancies probably decrease with rFSH compared to HMG/HP-HMG.

rFSH versus FSH-HP (23 studies, 4612 women)

- There is probably little to no difference in live births between rFSH and FSH-HP. If the chance of a live birth with FSH-HP is 27%, then the chance of a live birth with rFSH is between 24% and 31%.
- There is probably little to no difference in OHSS with rFSH compared to FSH-HP. If the risk of OHSS with FSH-HP is 2.5%, then the risk of OHSS with rFSH is between 1.7% and 3.8%.
- There is probably little to no difference in clinical pregnancy between rFSH and FSH-HP.

rFSH versus follitropin delta (7 studies, 3614 women)

- There is probably little or no difference in live births between rFSH and follitropin delta. If the chance of achieving a live birth with follitropin delta is 29%, the chance of a live birth with rFSH is between 24% and 30%.
- OHSS is probably higher with rFSH. If the risk of OHSS with follitropin delta is 4.4%, then the risk with rFSH is between 5.1% and 9.2%.
- There is probably little or no difference in clinical pregnancy between rFSH and follitropin delta.

rFSH versus biosimilars (8 studies, 2870 women)

- Live birth is probably higher with rFSH compared to biosimilars. If the chance of achieving a live birth with biosimilars is 23%, the chance of a live birth with rFSH is between 25% and 32%.
- There may be little or no difference in OHSS between rFSH and biosimilars. If the chance of OHSS with biosimilars is 9.6%, the chance of OHSS with rFSH is between 5.7% and 10%.
- Clinical pregnancy is probably higher with rFSH compared to biosimilars.

What are the limitations of the evidence?

The results mainly reflect IVF cycles with the transfer of a fresh embryo, so the results may be different for frozen embryos. Many studies were sponsored by pharmaceutical companies that manufacture the hormones in question, which could have affected the results. Furthermore, although we excluded some studies by using an integrity checklist, we have some concerns about the trustworthiness of the data from several of the studies.

How up to date is the evidence?

This updates a review previously published in 2011. The evidence is current to March 2025.

SUMMARY OF FINDINGS

Summary of findings 1. Summary of findings table - Recombinant follicle-stimulating hormone (rFSH) compared to human menopausal gonadotropin (HMG) or highly purified HMG (HP-HMG) for ovarian stimulation in assisted reproductive technology cycles

Recombinant follicle-stimulating hormone (rFSH) compared to human menopausal gonadotropin (HMG) or highly purified HMG (HP-HMG) for ovarian stimulation in assisted reproductive technology cycles

Patient or population: women undergoing in vitro fertilisation or intra-cytoplasmic sperm injection

Setting: clinic

Intervention: rFSH

Comparison: HMG/HP-HMG

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with HMG/HP-HMG	Risk with rFSH				
Live birth (or ongoing pregnancy)	279 per 1000	243 per 1000 (220 to 268)	OR 0.83 (0.73 to 0.95)	4793 (15 RCTs)	⊕⊕⊕⊖ Moderate ^a	
Ovarian hyperstimulation syndrome (OHSS)	26 per 1000	37 per 1000 (29 to 47)	OR 1.42 (1.12 to 1.80)	9813 (37 RCTs)	⊕⊕⊕⊖ Moderate ^a	
Clinical pregnancy	306 per 1000	278 per 1000 (251 to 302)	OR 0.87 (0.76 to 0.98)	4839 (15 RCTs)	⊕⊕⊕⊖ Moderate ^a	

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **OR:** odds ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradeapro.org/presentations/#/isof/isof_question_revman_web_465275035233290483.

^a Downgraded once for risk of bias -unclear risk of selective reporting; high risk of other bias

Summary of findings 2. Summary of findings table - Recombinant follicle-stimulating hormone (rFSH) compared to highly purified urinary follicle-stimulating hormone (FSH-HP) for ovarian stimulation in assisted reproductive technology cycles

Recombinant follicle-stimulating hormone (rFSH) compared to highly purified urinary follicle-stimulating hormone (FSH-HP) for ovarian stimulation in assisted reproductive technology cycles

Patient or population: women undergoing in vitro fertilisation or intra-cytoplasmic sperm injection

Setting: clinic

Intervention: rFSH

Comparison: FSH-HP

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with FSH-HP	Risk with rFSH				
Live birth (or ongoing pregnancy)	271 per 1000	275 per 1000 (242 to 310)	OR 1.02 (0.86 to 1.21)	2956 (15 RCTs)	⊕⊕⊕⊖ Moderate ^a	
Ovarian hyperstimulation syndrome (OHSS)	25 per 1000	26 per 1000 (17 to 38)	OR 1.01 (0.67 to 1.53)	3640 (18 RCTs)	⊕⊕⊕⊖ Moderate ^a	
Clinical pregnancy	336 per 1000	336 per 1000 (308 to 366)	OR 1.00 (0.88 to 1.14)	4535 (22 RCTs)	⊕⊕⊕⊖ Moderate ^a	

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

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See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_465275476103660607.

^a Downgraded once for risk of bias -unclear risk of selection bias and/or selective reporting; high risk of other bias

Summary of findings 3. Summary of findings table - Recombinant follicle-stimulating hormone (rFSH) compared to follitropin delta for ovarian stimulation in assisted reproductive technology cycles

Recombinant follicle-stimulating hormone (rFSH) compared to follitropin delta for ovarian stimulation in assisted reproductive technology cycles

Patient or population: women undergoing in vitro fertilisation or intra-cytoplasmic sperm injection

Setting: clinic

Intervention: rFSH

Comparison: follitropin delta

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with follitropin delta	Risk with rFSH				
Live birth (or ongoing pregnancy)	295 per 1000	269 per 1000 (236 to 303)	OR 0.88 (0.74 to 1.04)	2688 (3 RCTs)	⊕⊕⊕⊖ Moderate ^a	
Ovarian hyperstimulation syndrome (OHSS)	44 per 1000	69 per 1000 (51 to 92)	OR 1.62 (1.18 to 2.23)	2988 (4 RCTs)	⊕⊕⊕⊖ Moderate ^b	
Clinical pregnancy	325 per 1000	311 per 1000 (278 to 348)	OR 0.94 (0.80 to 1.11)	2688 (3 RCTs)	⊕⊕⊕⊕ High	

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **OR:** odds ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

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See interactive version of this table: https://gdt.gradeapro.org/presentations/#/isof/isof_question_revman_web_464822209368984499.

^a Downgraded once for inconsistency (I² = 55%)

^b Downgraded once for imprecision (small number of events)

Summary of findings 4. Summary of findings table - Recombinant follicle-stimulating hormone (rFSH) compared to biosimilars for ovarian stimulation in assisted reproductive technology cycles

Recombinant follicle-stimulating hormone (rFSH) compared to biosimilars for ovarian stimulation in assisted reproductive technology cycles

Patient or population: women undergoing in vitro fertilisation or intra-cytoplasmic sperm injection

Setting: clinic

Intervention: rFSH

Comparison: biosimilars

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with biosimilars	Risk with rFSH				
Live birth (or ongoing pregnancy)	233 per 1000	282 per 1000 (249 to 319)	OR 1.29 (1.09 to 1.54)	2883 (8 RCTs)	⊕⊕⊕⊖ Moderate ^a	
Ovarian hyperstimulation syndrome (OHSS)	96 per 1000	76 per 1000 (57 to 101)	OR 0.77 (0.57 to 1.06)	2883 (8 RCTs)	⊕⊕⊖⊖ Low ^{a,b}	
Clinical pregnancy	267 per 1000	329 per 1000 (293 to 368)	OR 1.35 (1.14 to 1.60)	2783 (7 RCTs)	⊕⊕⊕⊖ Moderate ^a	

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **OR:** odds ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

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See interactive version of this table: https://gdt.gradeapro.org/presentations/#/isof/isof_question_revman_web_46482222497680344.

^a Downgraded once for risk of bias - unclear risk of selection bias and/or selective reporting; high risk of other bias

^b Downgraded once for imprecision - small number of events

BACKGROUND

Description of the condition

Infertility is defined by the inability to achieve a pregnancy after 12 months of regular unprotected sexual intercourse. Common causes of infertility are tubal factor, male factor and unexplained infertility, amongst others.

Women who have a low chance of conceiving naturally may be treated with in vitro fertilisation (IVF) or intracytoplasmic sperm injection (ICSI). The first phase of IVF or ICSI consists of ovarian hyperstimulation to produce multiple follicles for follicle aspiration and subsequent fertilisation. A well-recognized complication of IVF or ICSI is ovarian hyperstimulation syndrome (OHSS).

Description of the intervention and how it might work

In the follicular phase of a normal menstrual cycle, a cohort of 10 to 20 ovarian antral follicles develops. Of this cohort, only one follicle obtains dominance over the others and shows continued growth until ovulation takes place. The aim in standard IVF or ICSI is to achieve the maturation of a much larger part of the ovarian antral follicle cohort. This is accomplished by ovarian stimulation with different types of hormones called gonadotropins.

The first generation of gonadotropins, used in the 1970s, was human menopausal gonadotropin (HMG). Produced from the urine of menopausal women, HMG is a combination of follicle stimulating hormone (FSH) and luteinising hormone (LH) in a 1:1 ratio. Since the 1980s, a variety of urinary gonadotropins have been produced, such as purified FSH (FSH-P), which contains less than 1 international unit (IU) of LH per 75 IU of FSH. The third generation of urinary gonadotropins was highly purified FSH (FSH-HP) with less than 0.1 IU of LH per 75 IU of FSH.

The fourth generation of gonadotropins was produced using recombinant DNA technology (recombinant FSH (rFSH) or follitropin alpha) using a Chinese hamster ovary cell line, which is free from LH activity. The production of rFSH is independent of urine collection, thus guaranteeing a high availability of a biochemically pure FSH preparation that is free from urinary protein contaminants. The production process also yields FSH with minimal batch-to-batch discrepancy and low immunogenicity, which allows subcutaneous administration [1].

In addition, rFSH generated from a human retinal cell line was developed (follitropin delta), which mainly differs from follitropin alpha in terms of a distinct glycosylation pattern [2, 3]. The dosing of follitropin delta is based on a personalised algorithm using a woman's anti-Müllerian hormone (AMH) level and body weight.

In recent years, multiple biosimilar drugs, which are highly similar to the original rFSH, have been developed. They are assessed on their pharmaceutical characteristics, efficacy and safety before introduction into the medical market. In the past decade, several biosimilars have been introduced into the pharmaceutical market in different countries [4, 5, 6, 7, 8, 9, 10, 11].

Why it is important to do this review

Given the availability of multiple types of gonadotropins for ovarian stimulation, it is important to examine whether there are differences in effectiveness and safety in the use of these

gonadotropins for IVF or ICSI. With this review, we aimed to bring together all randomised studies in this field under common inclusion criteria with consistent and valid statistical methods. This review is an update of Van Wely 2011 [12], which superseded the Cochrane reviews Van Wely 2003 [13], Daya 2000 [14], and Daya 2003 [15].

Apart from the different comparisons, two aspects in particular deserve attention. Firstly, different protocols are being used to achieve pituitary downregulation using either gonadotropin-releasing hormone (GnRH) agonists or GnRH antagonists, which may affect the outcome of assisted reproductive treatment cycles [16]. Secondly, many studies have been performed or sponsored by pharmaceutical companies, which may have introduced bias due to a conflict of interest [17].

OBJECTIVES

To compare the effectiveness and safety of recombinant follicle-stimulating hormone (rFSH) with the three main types of urinary gonadotropins, rFSH derived from a human cell line (follitropin delta) and rFSH biosimilars for ovarian stimulation in women undergoing IVF or ICSI treatment cycles.

METHODS

We followed the Methodological Expectations for Cochrane Intervention Reviews (MECIR) standards for conducting systematic reviews [18] and the PRISMA 2020 guideline for reporting systematic reviews [19].

Differences between protocol and review

We changed the title of the review, since additional comparisons were added to this update. We added the comparisons rFSH biosimilar and rFSH derived from a human cell line (follitropin delta).

We reported the comparisons with each urinary gonadotropin separately rather than in subgroups, as it was not our intention to compare those with each other.

We implemented the core outcome set in infertility [20]. We also added two outcomes: number of embryos, and number of embryos frozen.

We added cumulative live birth to the outcomes, thereby superseding the stratification of fresh cycles versus fresh plus frozen-thawed cycles from the previous version of this review. Regarding OHSS, there was no relevant cumulative outcome to report since none of the included studies used a freeze-all strategy.

We applied a checklist to assess trustworthiness in the included studies (TRACT checklist) [21].

We presented multiple pregnancies per woman randomised only.

We moved previously included abstracts that were still unpublished to 'awaiting classification', as we could not sensibly use the TRACT checklist on them, nor could we make sure outcomes were completely presented. For the same reasons, we moved all newly identified abstracts to 'awaiting classification'.

We presented the differences between the protocol and 2011 review in [Supplementary material 8](#).

Criteria for considering studies for this review

Types of studies

We included only randomised controlled trials (RCTs). We excluded quasi-RCTs, in which allocation was, for example, by alternation or reference to case record number or to dates of birth. We excluded cross-over trials, since the design was not appropriate in this context [22].

We included trials in which fresh and frozen embryos were transferred. We excluded trials with donor oocytes. We included individualised dose trials in the meta-analysis, but included dose-response trials only in the narrative review. We excluded studies investigating long-acting FSH (corifollitropin alfa), which is the topic of another Cochrane review [23].

Types of participants

Normogonadotrophic (defined as having normal serum concentration of FSH and LH) women undergoing fresh or frozen-thawed, or both, IVF or ICSI treatment cycles.

Types of interventions

Ovarian stimulation with rFSH versus:

- urinary gonadotropins (HMG, FSH-P or FSH-HP);
- rFSH derived from a human cell line (follitropin delta);
- biosimilar rFSH.

Outcome measures

Critical outcomes

Effectiveness

One or more live births per woman or, if not reported, one or more pregnancies ongoing beyond 20 weeks per woman.

Adverse events

Rate of ovarian hyperstimulation syndrome (OHSS) per woman.

Important outcomes

- Cumulative live birth or cumulative ongoing pregnancy per woman, including the result of frozen-thawed embryo transfers
- Clinical pregnancy – viable intrauterine pregnancy confirmed by ultrasound per woman
- Multiple pregnancies per woman
- Pregnancy loss per woman, including miscarriage, ectopic pregnancy, stillbirth and termination of pregnancy
- Gestational age at birth
- Birthweight
- Neonatal mortality
- Major congenital anomalies
- Patient acceptability/satisfaction

Process outcomes

- Amount of gonadotropins used per woman per cycle (total dose in IU)
- Duration of ovarian stimulation per woman per cycle
- Number of oocytes produced
- Number of embryos (as reported by study authors)

- Number of embryos frozen

Search methods for identification of studies

The search aimed to capture all published and unpublished RCTs that investigated the effectiveness of recombinant gonadotropins for ovarian stimulation, with guidance from the Cochrane Gynaecology and Fertility Information Specialist. We applied no language restrictions.

Electronic searches

We searched the following databases in March 2025.

- Cochrane Gynaecology and Fertility specialised register of controlled trials, ProCite platform, searched on 31 March 2025 for trials published up to 23 November 2023 (date of last trial entry into the specialised register);
- Cochrane Central Register of Controlled Trials (CENTRAL; 2025, Issue 3), via CRSO (now containing output from two trials registries and CINAHL), Web platform, searched 31 March 2025;
- MEDLINE, Ovid platform, searched from 1946 to 31 March 2025;
- Embase, Ovid platform, searched from 1980 to 31 March 2025.

There was no language restriction. Marian Showell, the Information Specialist of Cochrane Gynaecology and Fertility developed the search strategy. All search strategies are available in [Supplementary material 1](#).

We combined the MEDLINE search with the Cochrane Highly Sensitive Search Strategies for identifying randomised trials, which appear in Chapter 4 of the *Cochrane Handbook for Systematic Reviews of Interventions* [24]. We combined the Embase search strategy with an RCT filter developed by the Scottish Intercollegiate Guidelines Network (SIGN).

Searching other resources

Additionally, we checked all references in the published reviews, searched reference lists of the included studies, identified trials and background papers, and contacted study authors when necessary. Furthermore, we searched the trials registry clinicaltrials.gov (<https://clinicaltrials.gov/>) and the WHO International Clinical Trials Registry Platform (ICTRP) portal (<https://trialsearch.who.int/Default.aspx>) for additional (ongoing) trials. We also contacted pharmaceutical companies for any published, unpublished or ongoing studies not identified with our search strategy.

We also examined PubMed and Retraction Watch for any postpublication amendments to the included or eligible studies. These amendments may encompass expressions of concern, errata, corrigenda, or retractions.

Data collection and analysis

Selection of studies

Three authors (RPB, EBK, MvW) evaluated the results of the search strategy to identify potentially relevant trials, and retrieved the full articles using Covidence [25]. Review authors applied the selection criteria independently to the trial reports, rechecking trial eligibility and resolving disagreements by discussion with the other review authors. We excluded studies that did not meet the inclusion criteria and stated the reasons in the 'Characteristics of excluded studies' ([Supplementary material 3](#)). We screened all included

studies using the Trustworthiness in RANdomised Controlled Trials (TRACT) checklist [21]. We put studies that had been registered but not yet published, as well as potentially problematic studies with major concerns according to the integrity checklist, in 'awaiting classification' (Supplementary material 4).

Data extraction and management

Two review authors (RPB and EBK) independently extracted the outcome data and information on funding, location, clinical and design details and participants. We resolved any differences by discussion amongst the review authors. We entered details of the studies into the 'Characteristics of included studies' (Supplementary material 2). If data from the trial reports were insufficient or missing, we contacted the study authors for additional information. Where possible, we extracted data to allow an intention-to-treat analysis (the analysis should include all the participants in the groups to which they were originally randomly assigned).

Risk of bias assessment in included studies

Two review authors (RPB and EBK) independently extracted information regarding the risk of bias under seven domains (also see Supplementary material 9) using the Cochrane risk of bias assessment tool, RoB 1 [26]: selection bias (random sequence generation and allocation concealment); performance bias (blinding of participants and personnel); detection bias (blinding of outcome assessors); attrition bias (incomplete outcome data); reporting bias (selective reporting); and other bias. We assigned judgements as recommended by RoB 1 [26]. We resolved disagreements by discussion with a third review author (MvW). We described all judgements fully and presented the conclusions in the risk of bias table, which was incorporated into the interpretation of review findings by means of sensitivity analyses (see below). We rated the studies as at 'high', 'low' or 'unclear' risk of bias in each domain. We judged studies assessed with TRACT as having 'some concerns' to be at high risk of other bias.

Measures of treatment effect

We performed statistical analyses according to the statistical guidelines recommended in the *Cochrane Handbook* [27].

We summarised all binary outcomes using odds ratio (OR) with 95% confidence intervals (CI).

Number of oocytes produced per cycle, gestational age at birth, birthweight, ordinal scales used in patient acceptability, amount of gonadotropin used and duration of ovarian stimulation were treated as continuous outcomes. We abstracted, calculated or requested means and standard deviations.

Unit of analysis issues

All outcomes were expressed per woman randomised. Where only data 'per cycle' was available, and participants had contributed multiple started IVF cycles, we omitted data from the meta-analysis.

Dealing with missing data

Where there was insufficient information in the published report, we attempted to contact the study authors for clarification. If missing data became available, we included them in the analysis.

We anticipated that trials conducted before 2015 might not have data on live births of the study participants. We analysed data extracted from the trials on an intention-to-treat basis. Where randomised cases were missing from outcome assessment, we first contacted the study authors for additional data. If further data were not available, we assumed that missing participants had failed to achieve pregnancy and had not suffered reported adverse events.

Reporting bias assessment

When the paper measured but did not report data, we considered this to be internal reporting bias. To evaluate external reporting bias, we planned to perform funnel plots for the critical outcomes (primary analysis) and for clinical pregnancy if there were more than 10 studies.

Synthesis methods

We used Review Manager software to perform meta-analyses using a fixed-effect model [28]. For binary outcomes, we calculated the odds ratio. For reporting purposes, we also translated critical outcomes to absolute risks. We combined results for continuous outcomes using mean difference. If studies had reported acceptability scores using different scales, we would have used the standardised mean difference.

Prospectively, we planned to undertake three different stratifications of the critical outcomes:

- different GnRH protocols (antagonist, long GnRHa, short/ultrashort GnRHa protocol, and no GnRHa);
- use of fresh or frozen-thawed embryos;
- different sponsors (commercial, non-commercial) [29].

Investigation of heterogeneity and subgroup analysis

We anticipated heterogeneity between the predetermined strata and discussed possible reasons. We determined the presence of statistical heterogeneity of treatment effect amongst trials using the I^2 statistic [30].

Equity-related assessment

We did not analyse equity in this review, but we performed an overall evaluation on the potential influence of equity.

Sensitivity analysis

We assessed the influence of risk of bias on effect size by restricting the sensitivity analysis to studies at low risk of bias for the critical outcomes (defined as low risk of selection bias).

When the I^2 statistic was greater than 50%, we repeated the analyses using a random-effects model to assess whether conclusions differed from the primary analyses.

Certainty of the evidence assessment

We prepared summary of findings tables using GRADEpro GDT [31]. We assessed the certainty of evidence using GRADE methodology [32, 33].

The summary of findings tables evaluated the overall certainty of the body of evidence for the main review outcomes: live birth or ongoing pregnancy, OHSS and clinical pregnancy for the following review comparisons:

- rFSH versus HMG/HP-HMG;
- rFSH versus FSH-P (presented in a supplementary file as the intervention is no longer used);
- rFSH versus FSH-HP;
- rFSH human cell line (follitropin delta); and
- rFSH versus biosimilars.

Two review authors independently assessed the certainty of the evidence (RBS and EBK) using GRADE criteria: risk of bias, consistency of effect, imprecision, indirectness and publication bias. We made judgements about evidence certainty (high, moderate, low or very low) and resolved disagreements by discussion with a third review author (MvW). We justified our judgements, documented them, and incorporated them into the reporting of results for each outcome.

Consumer involvement

We did not involve consumers in the review development process.

RESULTS

Description of studies

For details on the studies, please see: [Supplementary material 2](#); [Supplementary material 3](#), [Supplementary material 4](#).

Results of the search

After removing duplicates, we screened 4249 records at abstract stage. We excluded 4120 records as irrelevant, leaving us with 129 records to screen at full-text stage. Of these, we excluded 55 studies (56 records) with reasons; we identified 17 studies as awaiting classification (19 records in total) and seven studies as ongoing. Finally, we included 25 new studies (45 records). Two of the newly identified records were clinical trials registry records of already included studies [34, 35]. For details of the article screening and selection process, see [Figure 1](#). Details of ongoing studies are described in [Supplementary material 5](#).

Figure 1. Study selection process

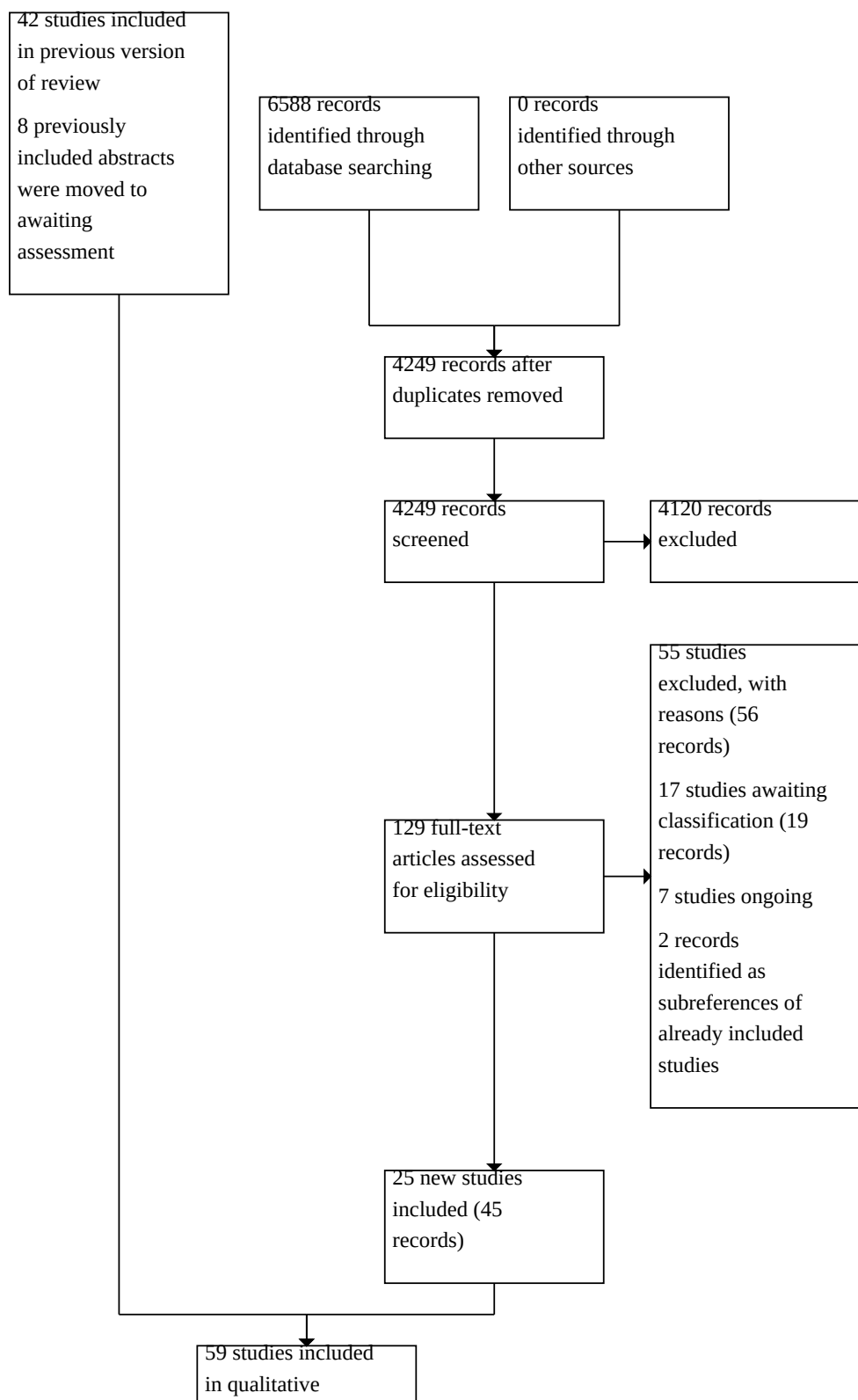
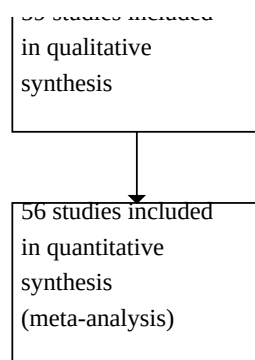


Figure 1. (Continued)



Included studies

In this update (2025), we have included 25 new studies (Aboulghar 2010 [36, 37]; Arce 2014 [38, 39]; Barakhoeva 2019 [4, 40, 41]; Chapon 2021 [42, 43]; Devroey 2012 [44]; Gholami 2010 [45]; Griesinger 2020 [46]; Hu 2020 [6, 47]; Hu 2023 [5, 48]; Ishihara 2021a [49]; Ishihara 2021b [50, 51]; Liu 2015 [52]; Moon 2007 [7]; NCT00257556 [53, 54]; NCT01687712 [9]; EUCTR2021-001785-38-ES [55, 56, 57]; Nyboe Andersen 2017 [58, 59, 60, 61, 62, 63]; Pasqualini 2021 [10]; Qiao 2021 [64, 65, 66]; Rettenbacher 2015 [8, 67]; Selman 2010 [68]; Sohrabvand 2012 [69]; Strowitzki 2016 [11]; Witz 2020 [70, 71, 72]; Ye 2012 [73]) from 45 records. See [Table 1](#).

The previous review, published in 2011, included 42 studies [12]. Eight of these were conference abstracts, which we have now placed in 'awaiting assessment'. Thus, the 2025 review update contains 59 included studies.

In total, the studies included 18,119 women. Details and characteristics of the included participants can be found in the Characteristics of included studies ([Supplementary material 2](#)).

The included studies varied in initial gonadotropin dose given, fertilisation method (IVF or ICSI), fresh or frozen cycles, outcome measurements, and (commercial) sponsorship ([Supplementary material 2](#)).

Comparisons

rFSH versus HMG or HP-HMG

Seventeen of the 59 included studies compared rFSH with HMG or HP-HMG (5639 women). The size of these studies varied from 40 women to 749 women. These 17 studies used the following down-regulation protocols:

- antagonist (5 studies; Bosch 2008 [35, 74], Chapon 2021; Devroey 2012; NCT00257556; Witz 2020);
- long GnRH agonist (9 studies; Andersen 2006 [75, 76]; Balasch 2003 [77]; European and Israeli Study Group 2002 [78]; Gordon 2001 [79]; Hompes 2008 [80]; Kilani 2003 [81]; Ng 2001 [82]; Westergaard 2001 [83]; Ye 2012);
- short GnRH agonist (2 studies; Rashidi 2005 [84]; Strehler 2001 [85]); and
- no down-regulation (1 study; Jansen 1998 [86]).

rFSH versus FSH-P

Five of the 59 included studies compared rFSH with FSH-P (1384 women, excluding Gordon 2001 (see below)). The size of these studies varied from 40 to 721 women. These five studies used the following down-regulation protocols:

- long GnRH agonist (4 studies; Gordon 2001; Hedon 1995 [87]; Out 1995 [88]; Recombinant human FSH Study Group 1995 [89]); and
- no down-regulation (1 study; Meden-Vrtovec 2003 [90]).

To prevent double counting of rFSH participants, we included Gordon 2001 only in the HMG/HP-HMG subgroup when reporting data per urinary gonadotropin separately.

rFSH versus FSH-HP

Twenty-three of the 59 included studies compared rFSH with FSH-HP (4612 women). The size of these studies varied from 65 to 486 women. These 23 studies used the following down-regulation protocols:

- long GnRH agonist (21 studies; Aboulghar 2010; Abate 2009 [91]; Antoine 2007 [34, 92]; Baker 2009 [93]; Bergh 1997 [94]; Dickey 2002 [95]; Dickey 2003 [96, 97]; Drakakis 2002 [98]; Franco 2000 [99]; Frydman 2000 [100]; Gallego 2003 [101]; Gholami 2010; Hoomans 1999 [102]; Lenton 2000 [103]; Liu 2015; Mohamed 2006 [104]; Nardo 2000 [105]; Schats 2000 [106]; Selman 2002 [107]; Selman 2010; Sohrabvand 2012);
- short GnRH agonist (2 studies; Cheon 2004 [108]; Hugues 2001 [109]).

rFSH versus follitropin delta

Seven of the 59 included studies compared rFSH with follitropin delta (3614 women). The size of these studies varied from 300 to 1326 women. All of these studies used GnRH antagonists for down regulation (Arce 2014; Griesinger 2020; Ishihara 2021a; Ishihara 2021b; Nyboe Andersen 2017; Qiao 2021; EUCTR2021-001785-38-ES).

Of these seven studies, three studies compared rFSH with rhFSH using a dose-response method, and we did not include them in the meta-analysis (Arce 2014; Griesinger 2020; Ishihara 2021b).

rFSH versus biosimilars

Eight of the 59 included studies compared rFSH with biosimilars (2870 women). The size of these studies varied from 86 to 1100 women. These eight studies used the following down-regulation protocols:

- long GnRH agonist (5 studies; Hu 2020; Hu 2023; NCT01687712; Rettenbacher 2015; Strowitzki 2016); and
- GnRH antagonist (3 studies; Barakhoeva 2019; Moon 2007; Pasqualini 2021).

Participants and setting

Participants were couples undergoing IVF or ICSI treatment. Most studies used single fresh transfer cycle. No studies performed freeze-all. All studies took place in a clinical setting.

Funding

Most studies were sponsored by the pharmaceutical industry. From the 59 included studies, 21 studies had either no or unknown sponsorship from a pharmaceutical company (Abate 2009; Balasch 2003; Bosch 2008; Chapon 2021; Cheon 2004; Drakakis 2002; Gallego 2003; Gholami 2010; Gordon 2001; Hugues 2001; Kilani 2003; Liu 2015; Meden-Vrtovec 2003; Mohamed 2006; Nardo 2000; Ng 2001; Rashidi 2005; Selman 2002; Sohrabvand 2012; Strehler 2001; Ye 2012).

Outcomes

For the critical outcomes, 45 studies reported on live birth (or ongoing pregnancy) and 49 studies reported on OHSS. Six studies reported on cumulative live birth (Andersen 2006; Devroey 2012; Hompes 2008; Out 1995; Strowitzki 2016; Witz 2020).

Excluded studies

We excluded 55 studies (56 records) in the 2025 review update (Abyholm 2008 [110]; Alama 2010 [111]; Arce 2020 [112, 113]; Bassil 2000 [114]; Boostanfar 2010 [115]; Boostanfar 2015 [116]; Bosch 2024 [117, 118]; Check 1995 [119]; Check 2001 [120]; ChiCTR-OPR-15006204 [121]; ChiCTR2000038030 [122]; CTRI/2017/11/010317 [123]; CTRI/2020/11/029333 [124]; CTRI/2022/03/040842 [125]; Daya 1995 [126]; Daya 1998 [127]; Edelstein 1990 [128]; EUCTR2013-003788-67-BE [129]; EUCTR2014-005099-27-ES [130]; EUCTR2015-002214-77-ES [131]; EUCTR2015-003779-31-ES [132]; EUCTR2015-005660-42-ES [133]; Filicori 2003 [134]; Frattarelli 2007 [135]; Fries 1990 [136]; Gocial

2000 [137]; Gomez 1999 [138]; Hillensjo 2009 [139]; Irvine 2000 [140]; Ismail 2002 [141]; Khair 2020 [142]; Kontic 1997 [143]; Kurji 2019 [144]; Lavy 1988 [145]; Lloyd 2003 [146]; Mahmoud 2001 [147]; Marrs 2003 [148]; Marshall 2002 [149]; Mayenga 1997 [150]; Melo 2010 [151]; Miron 1998 [152]; Mousavi 2011 [153]; Nakagawa 2007 [154]; NCT03071172 [155]; NCT03535103 [156]; NCT05266924 [157]; Platteau 2003 [158]; Scoccia 1987 [159]; Shao 2023 [160]; Soderstrom 1996 [161]; Tanbo 1990 [162]; Tulppala 1999 [163]; Wdowiak 2015 [164]; Westerguard 1996 [165]; Witz 2024 [166]).

We excluded the studies for the following reasons:

- ineligible intervention (31 studies);
- ineligible comparator (4 studies);
- ineligible population (4 studies);
- ineligible study design (12 studies);
- ineligible article type (3 studies)
- study terminated by the sponsor (1 study).

See [Supplementary material 3](#).

Studies awaiting classification

We placed 17 newly identified studies in awaiting classification (19 records): Aydin 2012 [167, 168]; Bu 2024 [169]; Chen 2004 [170]; CTRI/2016/10/007367 [171]; Hossein Rashidi 2021 [172]; Liu 2005 [173]; Mayenga 2006 [174]; Miller 2013 [175]; Motan 2011 [176]; NCT01337531 [177]; NCT01922193 [178]; Fernandez Ponce 2021 [179]; Heiser 2021 [180]; NCT04773353 [181]; Parsanezhad 2017 [182, 183]; Soong 1999 [184]; Turkcapar 2013 [185]).

We also moved eight previously included conference abstracts to awaiting classification, due to the inability to fully assess the studies' risk of bias and trustworthiness using TRACT (Alvino 1995 [186]; Berger 1999 [187]; Ferraretti 1999 [188, 189]; Germond 2000 [190]; Ghosh 1999 [191, 192]; Kornilov 1999 [193]; Machado 1999 [194]; O'Dea 1993 [195]).

One study by Chakravarty [192] previously listed in awaiting classification is a subreference of Ghosh 1999 [191]. For the full details of the studies listed in awaiting classification, refer to [Supplementary material 4](#).

Risk of bias in included studies

The risk of bias judgements per included study per domain are described in [Supplementary material 2](#). Also, see [Figure 2](#).

Figure 2. Risk of bias summary: review authors' judgements about each risk of bias item for each included study

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias)	Blinding of outcome assessment (detection bias)	Incomplete outcome data (attrition bias): All outcomes	Selective reporting (reporting bias)	Other bias
Abate 2009	+	?	+	+	-	-	-
Aboulghar 2010	?	+	+	+	+	?	-
Andersen 2006	+	?	+	+	+	?	?
Antoine 2007	+	+	+	+	?	?	-
Arce 2014	+	+	+	+	+	?	-
Baker 2009	+	?	+	+	+	?	-
Balasch 2003	?	-	+	+	-	?	-
Barakhoeva 2019	+	+	+	+	+	?	-
Bergh 1997	+	+	+	+	+	?	-
Bosch 2008	+	+	+	+	+	?	-
Chapon 2021	+	?	+	+	+	?	-
Cheon 2004	?	?	+	+	?	-	-
Devroey 2012	+	+	+	+	+	?	-
Dickey 2002	+	?	+	+	+	?	-
Dickey 2003	+	?	+	+	+	?	-
Drakakis 2002	?	?	+	+	-	-	-
EUCTR2021-001785-38-ES	?	?	+	+	+	+	-
European and Israeli Study Group 2002	+	+	+	+	+	?	?
Franco 2000	?	?	+	+	+	?	-
Frydman 2000	+	?	+	+	+	?	-
Gallego 2003	+	?	+	+	+	?	-

Figure 2. (Continued)

Frydman 2000	+	?	+	+	+	?	+
Gallego 2003	+	?	+	+	+	?	-
Gholami 2010	?	?	+	+	+	?	-
Gordon 2001	+	+	+	+	+	?	-
Griesinger 2020	+	+	+	+	+	+	?
Hedon 1995	+	?	+	+	+	?	-
Hompes 2008	+	+	+	+	+	?	?
Hoomans 1999	+	+	+	+	+	?	-
Hu 2020	+	?	+	+	+	?	-
Hu 2023	+	+	+	+	+	+	-
Hugues 2001	+	?	+	+	+	?	-
Ishihara 2021a	+	+	+	+	+	?	?
Ishihara 2021b	+	+	+	+	+	?	?
Jansen 1998	+	?	+	+	+	?	-
Kilani 2003	+	?	+	+	+	?	-
Lenton 2000	+	+	+	+	+	?	-
Liu 2015	+	?	+	+	?	?	-
Meden-Vrtovec 2003	?	?	+	+	+	?	-
Mohamed 2006	+	?	+	+	+	-	-
Moon 2007	+	?	+	+	+	?	-
Nardo 2000	?	?	+	+	+	?	-
NCT00257556	?	?	+	+	+	?	-
NCT01687712	?	?	+	+	+	?	-
Ng 2001	+	?	+	+	+	?	-
Nyboe Andersen 2017	+	+	+	+	+	+	?
Out 1995	?	?	+	+	+	?	-
Pasqualini 2021	+	+	+	+	-	-	-
Qiao 2021	+	+	+	+	+	+	?
Rashidi 2005	+	?	+	+	+	?	-
Recombinant human FSH Study Group 1995	+	+	+	+	?	?	?
Rettenbacher 2015	?	+	+	+	-	+	?
Schats 2000	+	?	+	+	+	?	-
Selman 2002	+	?	+	+	+	?	-
Selman 2010	+	?	+	+	+	?	-
Sohrabvand 2012	+	?	+	+	+	?	-
Strehler 2001	+	+	+	+	+	?	-

Figure 2. (Continued)

Strehler 2001	+	+	+	+	+	?	—
Strowitzki 2016	+	+	+	+	+	?	?
Westergaard 2001	+	?	+	+	+	?	—
Witz 2020	+	+	+	+	+	?	—
Ye 2012	+	?	+	+	?	?	—

Random sequence generation

All 59 included studies were RCTs, but for 13 studies the method of randomisation remained unclear (Aboulghar 2010; Balasch 2003; Cheon 2004; Drakakis 2002; EUCTR2021-001785-38-ES; Franco 2000; Gholami 2010; Meden-Vrtovec 2003; Nardo 2000; NCT00257556; NCT01687712; Out 1995; Rettenbacher 2015). The rest of the studies were at low risk of bias in this domain as they adequately explained how the random sequence was generated.

Allocation concealment

One study was at high risk of bias for allocation concealment (Balasch 2003) and 34 studies at unclear risk of bias in this domain (Abate 2009; Andersen 2006; Baker 2009; Chapon 2021; Cheon 2004; Dickey 2002; Dickey 2003; Drakakis 2002; EUCTR2021-001785-38-ES; Franco 2000; Frydman 2000; Gallego 2003; Gholami 2010; Hedon 1995; Hu 2020; Hugues 2001; Jansen 1998; Kilani 2003; Liu 2015; Meden-Vrtovec 2003; Mohamed 2006; Moon 2007; Nardo 2000; NCT00257556; NCT01687712; Ng 2001; Out 1995; Rashidi 2005; Schats 2000; Selman 2002; Selman 2010; Sohrabvand 2012; Westergaard 2001; Ye 2012). We assessed the rest of the studies as being at low risk of bias in this domain as they adequately explained their allocation concealment method. See [Figure 2](#).

Blinding of participants and personnel (performance bias) and blinding of outcome assessment (detection bias)

We judged all included studies to be at low risk of bias in these domains as blinding was unlikely to have influenced the outcomes.

Incomplete outcome data (attrition bias)

For the pregnancy outcomes, studies presented data according to intention-to-treat (ITT) or otherwise, we could usually retrace the ITT data from additional information from the respective studies.

Studies did not usually report the amount of gonadotropins used, the duration of stimulation and the number of oocytes retrieved per woman randomised, but reported them in a subgroup of women that either received gonadotropins or a subgroup that underwent oocyte pick-up. This may have introduced bias, therefore we have not pooled these outcomes in meta-analyses for all comparisons.

We included the number of embryos and the number of embryos frozen in a narrative review only, due to high heterogeneity ([Table 2](#); [Table 3](#)).

We judged five studies to be at high risk of attrition bias (Abate 2009; Balasch 2003; Drakakis 2002; Pasqualini 2021; Rettenbacher 2015) and five studies at unclear risk of attrition bias (Antoine 2007; Cheon 2004; Liu 2015; Recombinant human FSH Study Group 1995; Ye 2012). The rest of the studies were at low risk of attrition bias.

Selective reporting (reporting bias)

Six studies were at low risk of reporting bias as they reported all prespecified outcomes (EUCTR2021-001785-38-ES; Griesinger 2020; Hu 2023; Nyboe Andersen 2017; Qiao 2021; Rettenbacher 2015). We judged five studies to be at high risk of reporting bias (Abate 2009; Cheon 2004; Drakakis 2002; Mohamed 2006; Pasqualini 2021). The rest of the studies were at unclear risk of reporting bias.

Other bias

Of the 59 studies, 38 were industry-sponsored. This is an important issue since there could be a conflict of interest.

- Seven studies were sponsored by Serono (Bergh 1997; Franco 2000; Frydman 2000; Hedon 1995; Lenton 2000; Schats 2000; Recombinant human FSH Study Group 1995).
- Three studies were sponsored by Organon (Hoomans 1999; Jansen 1998; Out 1995).
- Ten studies had various pharmaceutical sponsors:
 - IVFarma LLC (Barakhoeva 2019);
 - Glycotope GmbH (Griesinger 2020);
 - Qilu Pharmaceutical Co., Ltd (Hu 2023);
 - Dong-A Pharmaceuticals Co (Moon 2007);
 - Fertility Biotech AG (NCT01687712);
 - Gema Biotech S.A (Pasqualini 2021);
 - Finox AG (Rettenbacher 2015);
 - Praxi Provita (Selman 2010);
 - TEVA Pharmaceuticals (Strowitzki 2016);
 - Hangzhou Yuyuan Bioscience Technology Co., Ltd. (Hu 2020).
- Fifteen studies were sponsored by Ferring (Andersen 2006; Arce 2014; Devroey 2012; Dickey 2002; Dickey 2003; European and Israeli Study Group 2002; EUCTR2021-001785-38-ES; Hompes 2008; Ishihara 2021a; Ishihara 2021b; NCT00257556; Nyboe Andersen 2017; Qiao 2021; Westergaard 2001; Witz 2020).
- Three studies were sponsored by IBSA (Aboulghar 2010; Antoine 2007; Baker 2009).

Funding was unclear in six studies (Abate 2009; Drakakis 2002; Gallego 2003; Gholami 2010; Nardo 2000; Sohrabvand 2012).

A further fifteen studies reported having no funding or being funded by governmental funding (Balasch 2003; Bosch 2008; Chapon 2021; Cheon 2004; Gordon 2001; Hugues 2001; Kilani 2003; Liu 2015; Meden-Vrtovec 2003; Mohamed 2006; Ng 2001; Rashidi 2005; Selman 2002; Strehler 2001; Ye 2012).

We assessed all included studies using the trustworthiness checklist TRACT [21]. The TRACT assessment resulted in most studies having 'some concerns', thus we judged them as being at

high risk of other bias. The TRACT assessment resulted in 11 studies having 'no concerns' but, due to pharma funding, we judged them to be at unclear risk of bias (Andersen 2006; European and Israeli Study Group 2002; Griesinger 2020; Hompes 2008; Ishihara 2021a; Ishihara 2021b; Nyboe Andersen 2017; Qiao 2021; Rettenbacher 2015; Recombinant human FSH Study Group 1995; Strowitzki 2016).

Synthesis of results

Absolute and relative effects for the critical outcomes live birth and OHSS and for the important outcome clinical pregnancy are summarised with their GRADE assessment ([Summary of findings 1](#); [Summary of findings 2](#); [Summary of findings 3](#); [Summary of findings 4](#); [Supplementary material 10](#)).

1. rFSH versus HMG/HP-HMG

Data are presented in [Supplementary material 6](#) under 'Analysis group 1: rFSH versus HMG/HP-HMG' and [Summary of findings 1](#).

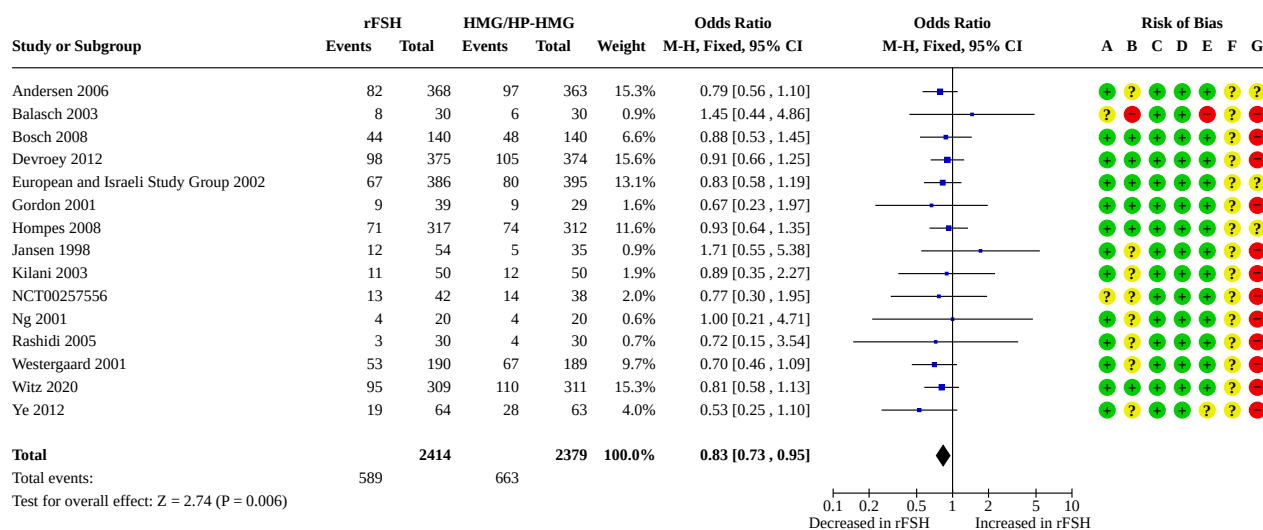
We included 15 studies for the comparison rFSH versus HMG/HP-HMG (Andersen 2006; Balasch 2003; Bosch 2008; Devroey 2012; European and Israeli Study Group 2002; Gordon 2001; Hompes 2008; Jansen 1998; Kilani 2003; NCT00257556; Ng 2001; Rashidi 2005; Westergaard 2001; Witz 2020; Ye 2012).

Critical outcomes

Live birth (or ongoing pregnancy)

Live birth is probably lower with rFSH compared to HMG/HP-HMG (OR 0.83, 95% CI 0.73 to 0.95; $I^2 = 0\%$; 15 studies, 4793 participants; [Figure 3](#), moderate-certainty evidence). If the chance of live birth with HMG/HP-HMG is 28%, then the chance of live birth with rFSH is between 22% and 27%.

Figure 3. Comparison rFSH versus HMG/HP-HMG - outcome live birth (or ongoing pregnancy)



Risk of bias legend

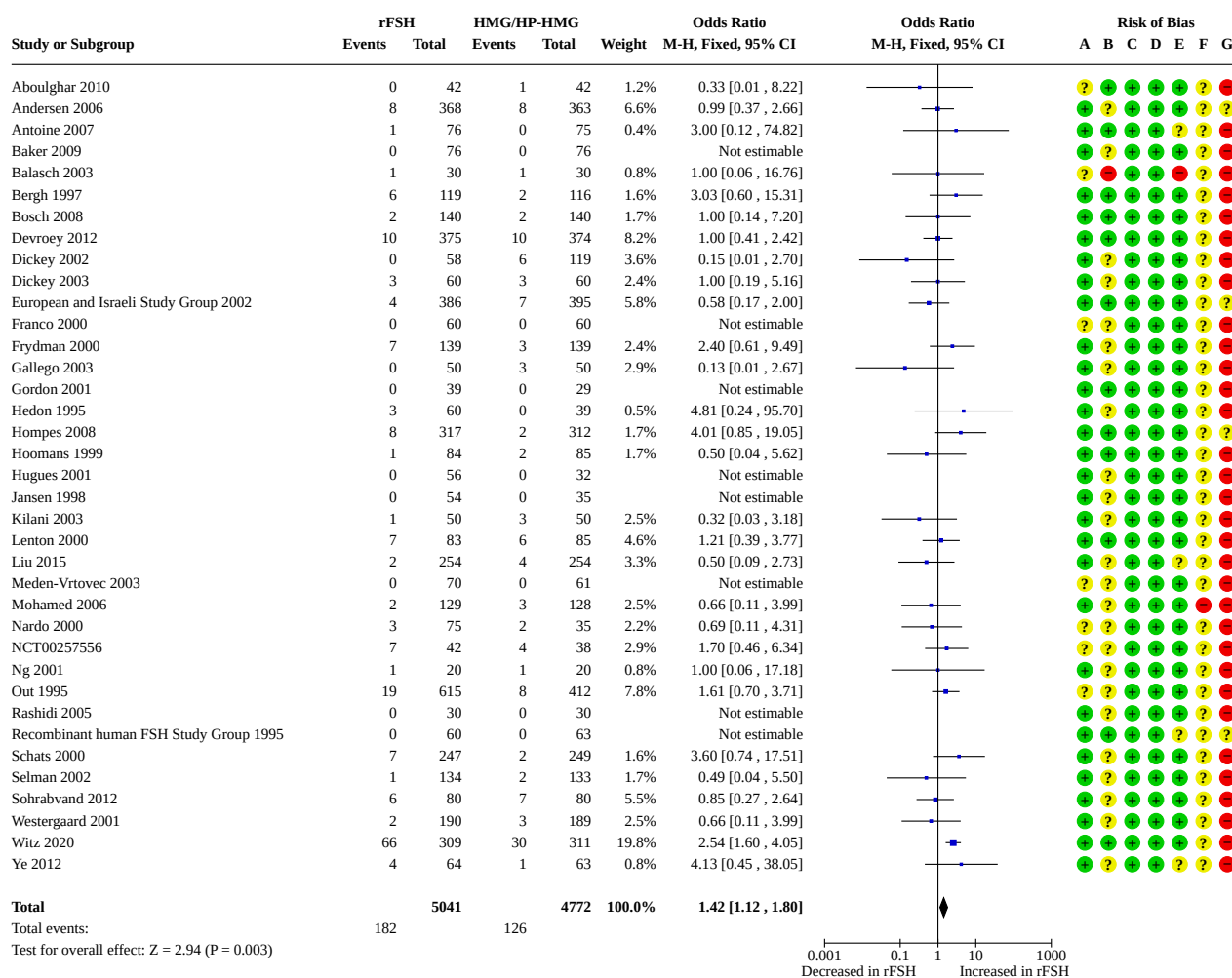
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

OHSS

OHSS is probably higher with rFSH when compared to HMG/HP-HMG (OR 1.42, 95% CI 1.12 to 1.80; $I^2 = 3\%$; 37 studies, 9813

participants; [Figure 4](#); moderate-certainty evidence). If the risk of OHSS with HMG/HP-HMG is 2.6%, then the risk of OHSS with rFSH is between 2.9% and 4.7%.

Figure 4. Comparison rFSH versus HMG/HP-HMG - outcome OHSS

**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

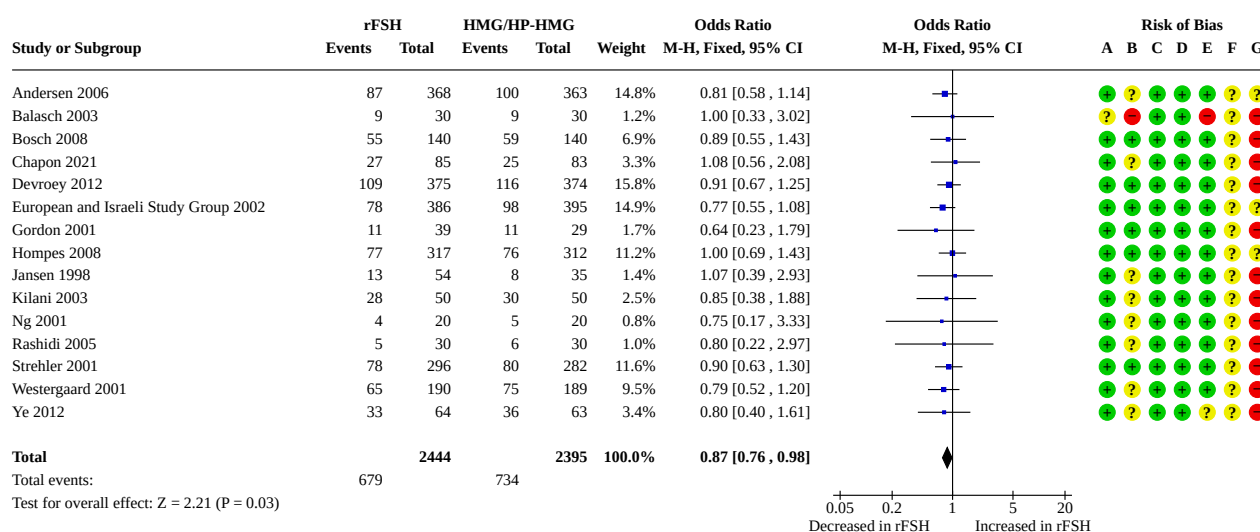
Important outcomes**Cumulative live birth (or cumulative ongoing pregnancy)**

There is probably little or no difference in cumulative live birth when comparing rFSH with HMG/HP-HMG (OR 0.92, 95% CI 0.78 to 1.08; I² = 0%; 4 studies, 2729 participants).

Clinical pregnancy

Clinical pregnancy is probably lower with rFSH compared to HMG/HP-HMG (OR 0.87, 95% CI 0.76 to 0.98; I² = 0%; 15 studies, 4839 participants; Figure 5; moderate-certainty evidence). If the chance of achieving a clinical pregnancy with HMG/HP-HMG is 31%, then the chance of a clinical pregnancy with rFSH is between 25% and 30%.

Figure 5. Comparison rFSH versus HMG/HP-HMG - outcome clinical pregnancy

**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Multiple pregnancy

Results suggested there may be a decrease in multiple pregnancy in rFSH compared to HMG/HP-HMG (OR 0.73, 95% CI 0.57 to 0.94; I² = 0%; 12 studies, 4476 participants).

Miscarriage

Results suggested there is probably little or no difference in miscarriage rate when comparing rFSH with HMG/HP-HMG (OR 1.08, 95% CI 0.84 to 1.39; I² = 0%; 15 studies, 4813 participants).

Ectopic pregnancy

We are uncertain whether there is a difference in ectopic pregnancy when comparing rFSH with HMG/HP-HMG (OR 1.99, 95% CI 0.08 to 50.04; 1 study, 109 participants).

Stillbirth

Results suggested there may be no difference in stillbirth when comparing rFSH to HMG/HP-HMG (OR 1.66, 95% CI 0.22 to 12.64; I² = 0%; 3 studies, 1760 participants).

Termination of pregnancy

When comparing rFSH to HMG/HP-HMG, there may be little or no difference in termination of pregnancy (OR 0.71, 95% CI 0.14 to 3.59; I² = 29%; 2 studies, 1480 participants).

Major congenital anomalies

We are uncertain whether there may be a difference in major congenital abnormalities when comparing rFSH to HMG/HP-HMG (OR 5.04, 95% CI 0.59 to 43.35; 1 study, 749 participants).

Patient acceptability/satisfaction

No studies in this comparison reported this outcome.

Amount of gonadotropins used

Due to high heterogeneity, we did not pool the result in meta-analysis (Analysis 1.11).

Duration of ovarian stimulation

Due to high heterogeneity, we did not pool the result in meta-analysis (Analysis 1.12).

Number of oocytes produced per cycle

Due to high heterogeneity, we did not pool the result in meta-analysis (Analysis 1.13).

Number of embryos (as reported by study authors)

Available results are presented in Table 2.

Number of embryos frozen

Available results are presented in Table 3.

Supporting analyses**Sensitivity analysis**

We carried out sensitivity analyses restricted to studies that we judged to be at low risk of selection bias.

Live birth (or ongoing pregnancy)

We included six studies for the comparison rFSH versus HMG/HP-HMG (Bosch 2008; Devroey 2012; European and Israeli Study Group 2002; Gordon 2001; Hompes 2008; Witz 2020). There is probably

little to no difference in live birth comparing rFSH with HMG/HP-HMG (OR 0.86, 95% CI 0.73 to 1.01; $I^2 = 0\%$; 6 studies, 3127 participants).

OHSS

We included six studies for the comparison rFSH versus HMG/HP-HMG (Bosch 2008; Devroey 2012; European and Israeli Study Group 2002; Gordon 2001; Hompes 2008; Witz 2020). As in the main analysis, OHSS is probably higher with rFSH compared to HMG/HP-HMG (OR 1.89, 95% CI 1.32 to 2.72; $I^2 = 52\%$; 6 studies, 3127 participants).

2. rFSH versus FSH-P

Data are presented in [Supplementary material 6](#) under 'Analysis group 2: rFSH versus FSH-P' and [Supplementary material 10](#).

We included four studies for the comparison rFSH versus FSH-P (Hedon 1995; Meden-Vrtovec 2003; Out 1995; Recombinant human FSH Study Group 1995).

Critical outcomes

Live birth (or ongoing pregnancy)

There may be little to no difference in live birth between rFSH and FSH-P (OR 1.22, 95% CI 0.92 to 1.60; $I^2 = 0\%$; 4 studies, 1380 participants, low-certainty evidence). If the chance of live birth with FSH-P is 18%, then the chance of live birth with rFSH is between 16% and 25%.

OHSS

There may be little to no difference in OHSS between rFSH and FSH-P (OR 1.79, 95% CI 0.81 to 3.99; $I^2 = 0\%$; 4 studies, 1380 participants, low-certainty evidence). If the risk of OHSS with FSH-P is 1.4%, then the risk of OHSS with rFSH is between 1.1% and 5.3%.

Important outcomes

Cumulative live birth (or cumulative ongoing pregnancy)

Cumulative live birth may be higher with rFSH versus FSH-P, but the confidence interval included no effect (OR 1.33, 95% CI 0.98 to 1.81; 1 study, 1027 participants).

Clinical pregnancy

There may be little or no difference in clinical pregnancy when comparing rFSH with FSH-P (OR 1.23, 95% CI 0.96 to 1.56; $I^2 = 0\%$; 5 studies, 1450 participants; low-certainty evidence). If the chance of clinical pregnancy with FSH-P is 23%, then the chance of clinical pregnancy with rFSH is between 22% and 32%.

Multiple pregnancy

There may be little or no difference in multiple pregnancy when comparing rFSH to FSH-P (OR 1.38, 95% CI 0.48 to 3.96; $I^2 = 0\%$; 2 studies, 222 participants).

Miscarriage

Results suggested there may be little or no difference in miscarriage rate when comparing rFSH with FSH-P (OR 2.14, 95% CI 0.19 to 24.21; 1 study, 123 participants).

Ectopic pregnancy

We are uncertain whether there is a difference in ectopic pregnancy when comparing rFSH with FSH-P (OR 3.20, 95% CI 0.13 to 80.14; 1 study, 123 participants).

Stillbirth

No studies that compared rFSH with FSH-P reported on stillbirth.

Termination of pregnancy

No studies that compared rFSH with FSH-P reported on termination of pregnancy.

Major congenital anomalies

We are uncertain whether there may be a difference in major congenital abnormalities when comparing rFSH to FSH-P (OR 0.34, 95% CI 0.01 to 8.62; 1 study, 123 participants).

Patient acceptability/satisfaction

No studies in this comparison reported this outcome.

Amount of gonadotropins used

Due to high heterogeneity, we did not pool the result in meta-analysis (Analysis 2.9).

Duration of ovarian stimulation

Due to high heterogeneity, we did not pool the result in meta-analysis (Analysis 2.10).

Number of oocytes produced per cycle

Due to high heterogeneity, we did not pool the result in meta-analysis (Analysis 2.11).

Number of embryos (as reported by study authors)

Available results are presented in [Table 2](#).

Number of embryos frozen

Available results are presented in [Table 3](#).

Supporting analyses

Sensitivity analyses

We carried out sensitivity analyses restricted to studies that we judged to be at low risk of selection bias.

Live birth (or ongoing pregnancy)

We included one study for the comparison rFSH versus FSH-P (Recombinant human FSH Study Group 1995). Similar to the main analysis, there may be little or no difference between the two groups (OR 1.21, 95% CI 0.43 to 3.38; 1 study, 123 participants).

OHSS

We included one study for the comparison rFSH versus FSH-P (Recombinant human FSH Study Group 1995). There was no incidence of OHSS in this study.

3. rFSH versus FSH-HP

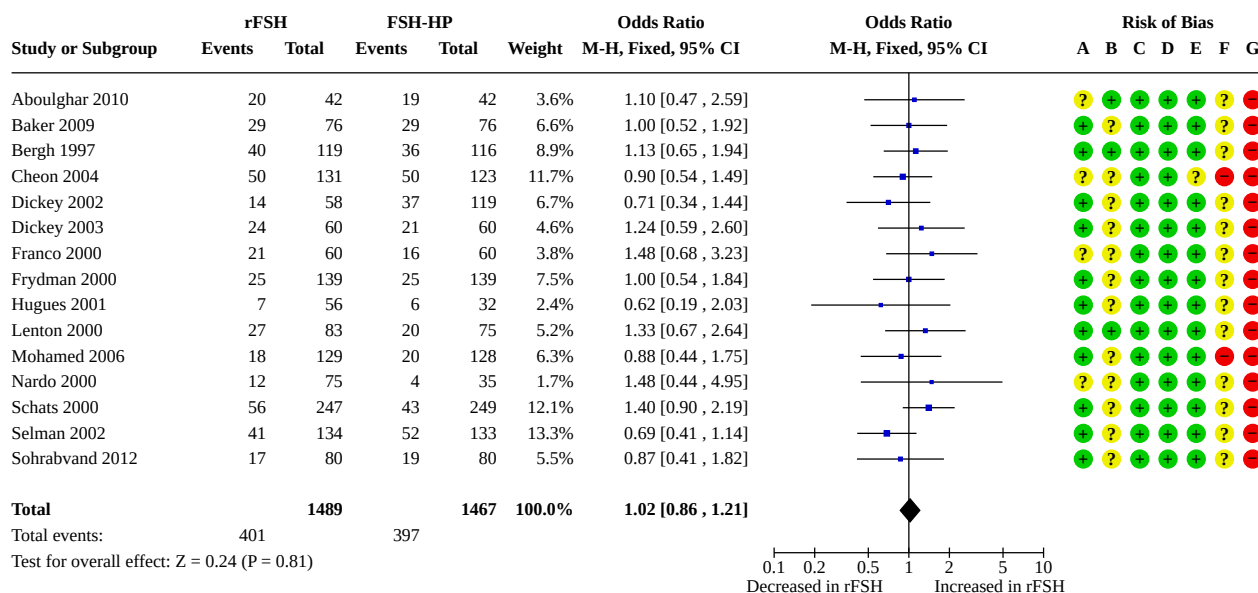
Data are presented in [Supplementary material 6](#) under 'Analysis group 3: rFSH versus FSH-HP' and [Summary of findings 2](#).

Critical outcomes**Live birth (or ongoing pregnancy)**

We included 15 studies in this outcome (Aboulghar 2010; Baker 2009; Bergh 1997; Cheon 2004; Dickey 2002; Dickey 2003; Franco 2000; Frydman 2000; Hugues 2001; Lenton 2000; Mohamed 2006;

Nardo 2000; Schats 2000; Selman 2002; Sohrabvand 2012). There is probably little to no difference in live birth between rFSH and FSH-HP (OR 1.02, 95% CI 0.86 to 1.21; $I^2 = 0\%$; 15 studies, 2956 participants; [Figure 6](#), moderate-certainty evidence). If the chance of live birth with FSH-HP is 27%, then the chance of live birth with rFSH is between 24% and 31%.

Figure 6. Comparison rFSH versus FSH-HP – outcome live birth (or ongoing pregnancy)

**Risk of bias legend**

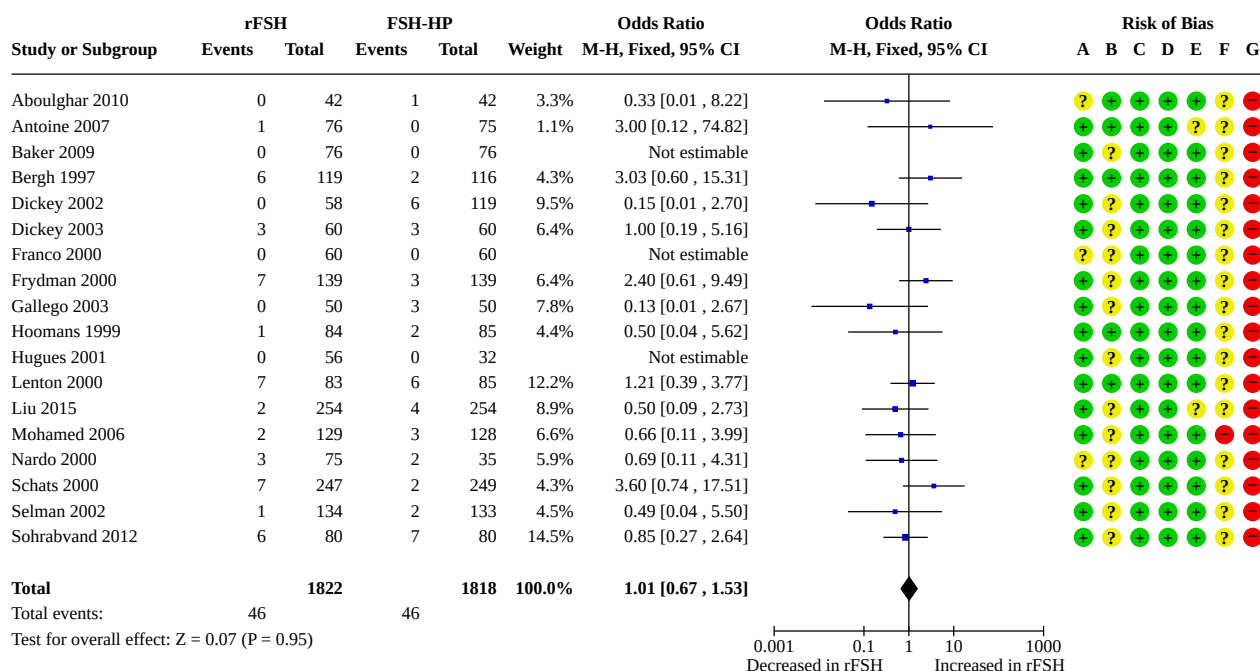
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

OHSS

We included 18 studies for the outcome OHSS (Aboulghar 2010; Antoine 2007; Baker 2009; Bergh 1997; Dickey 2002; Dickey 2003; Franco 2000; Frydman 2000; Gallego 2003; Hoomans 1999; Hugues 2001; Lenton 2000; Liu 2015; Mohamed 2006; Nardo 2000; Schats

2000; Selman 2002; Sohrabvand 2012). There is probably little to no difference in OHSS when comparing rFSH versus FSH-HP (OR 1.01, 95% CI 0.67 to 1.53; $I^2 = 0\%$; 18 studies, 3640 participants; [Figure 7](#), moderate-certainty evidence). If the risk of OHSS with FSH-HP is 2.5%, then the risk of OHSS with rFSH is between 1.7% and 3.8%.

Figure 7. Comparison rFSH versus FSH-HP - outcome OHSS

**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

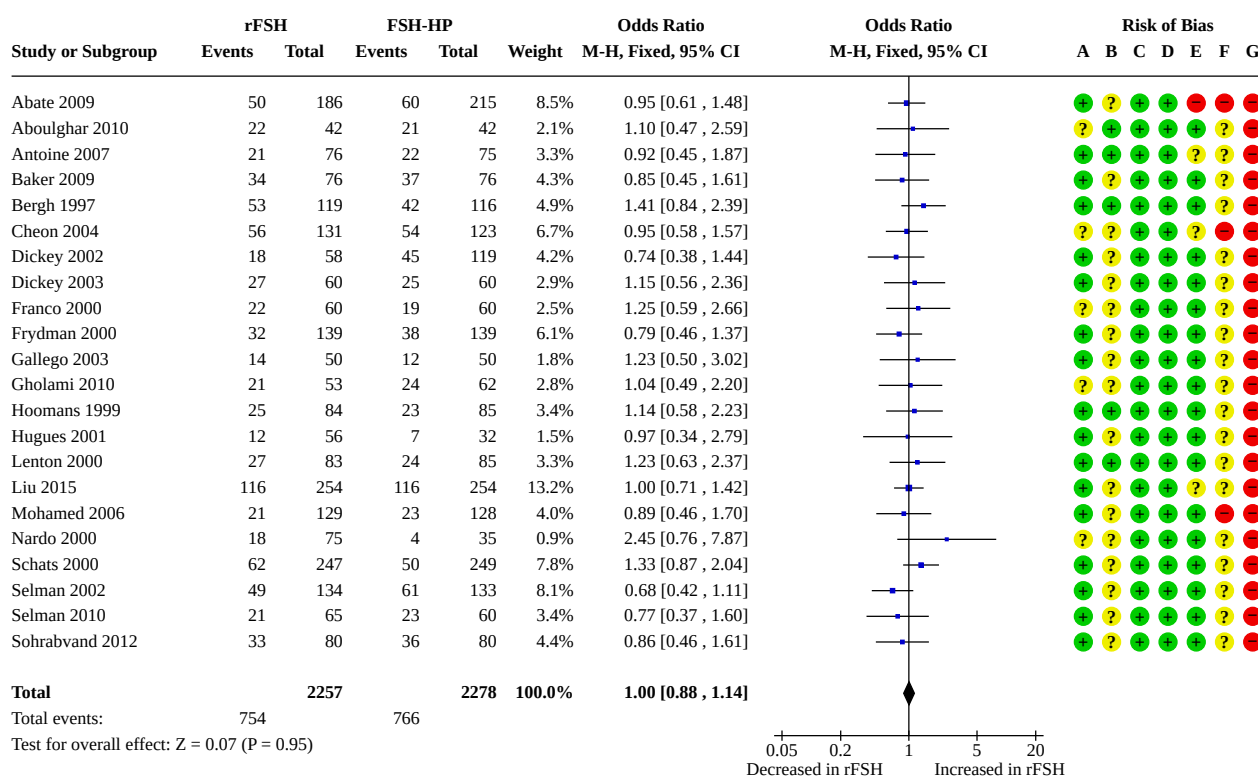
Important outcomes**Cumulative live birth (or cumulative ongoing pregnancy)**

No studies that compared rFSH with FSH-HP reported on cumulative live birth.

Clinical pregnancy

There is probably little to no difference in clinical pregnancy when comparing rFSH with FSH-HP (OR 1.00, 95% CI 0.88 to 1.14; $I^2 = 0\%$; 22 studies, 4535 participants; Figure 8; moderate-certainty evidence). If the chance of clinical pregnancy with FSH-HP is 34%, then the chance of clinical pregnancy with rFSH is between 31% and 37%.

Figure 8. Comparison rFSH versus FSH-HP - outcome clinical pregnancy



Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Multiple pregnancy

There may be little to no difference in multiple pregnancy when comparing rFSH to FSH-HP (OR 1.09, 95% CI 0.84 to 1.40; I² = 0%; 13 studies, 2526 participants).

Miscarriage

There may be little or no difference in miscarriage rate when comparing rFSH with FSH-HP (OR 1.22, 95% CI 0.97 to 1.54; I² = 0%; 20 studies, 3933 participants).

Ectopic pregnancy

There may be little to no difference in ectopic pregnancy when comparing rFSH with FSH-HP (OR 0.78, 95% CI 0.19 to 3.17; I² = 0%; 4 studies, 1051 participants).

Stillbirth

No studies that compared rFSH with FSH-HP reported on stillbirth.

Termination of pregnancy

No studies that compared rFSH with FSH-HP reported on termination of pregnancy.

Gestational age at birth

No studies that compared rFSH with FSH-HP reported on gestational age at birth.

Birthweight

No studies that compared rFSH with FSH-HP reported on birthweight.

Neonatal mortality

No studies that compared rFSH with FSH-HP reported on neonatal mortality.

Major congenital anomalies

No studies that compared rFSH with FSH-HP reported on major congenital anomalies.

Patient acceptability/satisfaction

No studies that compared rFSH with FSH-HP reported on patient acceptability/satisfaction.

Amount of gonadotropins used

Due to high heterogeneity, we did not pool the result in meta-analysis (Analysis 3.7).

Duration of ovarian stimulation

Due to high heterogeneity, we did not pool the result in meta-analysis (Analysis 3.8).

Number of oocytes produced per cycle

Due to high heterogeneity, we did not pool the result in meta-analysis (Analysis 3.9).

Number of embryos (as reported by study authors)

Available results are presented in [Table 2](#).

Number of embryos frozen

Available results are presented in [Table 3](#).

Supporting analyses

Sensitivity analyses

We carried out sensitivity analyses restricted to studies that we judged to be at low risk of selection bias.

Live birth (or ongoing pregnancy)

We included two studies for the comparison rFSH versus FSH-HP (Bergh 1997; Lenton 2000). Similar to the main analysis, there is probably little or no difference between the two groups (OR 1.20, 95% CI 0.78 to 1.84; $I^2 = 0\%$; 2 studies, 393 participants).

OHSS

We included four studies for the comparison rFSH versus FSH-HP (Antoine 2007; Bergh 1997; Hoomans 1999; Lenton 2000). As in the main analysis, there is probably little or no difference in OHSS between rFSH and FSH-HP (OR 1.52, 95% CI 0.68 to 3.39; $I^2 = 0\%$; 4 studies, 723 participants).

4. Supporting analyses: rFSH versus urinary gonadotropins – grouped by down-regulation protocol and by sponsor

Data are presented in [Supplementary material 6](#) under 'Analysis group 4: rFSH versus urinary gonadotropins – by down-regulation protocol and by sponsor (critical outcomes)'.

Live birth (or ongoing pregnancy) by down-regulation protocol

Of the 34 studies that reported on live birth:

- four used an antagonist protocol (Bosch 2008; Devroey 2012; NCT00257556; Witz 2020);
- 25 used a long GnRH agonist protocol (Aboulghar 2010; Sohravand 2012; Andersen 2006; Baker 2009; Balasch 2003; Bergh 1997; Dickey 2002; Dickey 2003; European and Israeli Study Group 2002; Franco 2000; Frydman 2000; Gordon 2001; Hedon 1995; Hompes 2008; Kilani 2003; Lenton 2000; Mohamed 2006; Nardo 2000; Ng 2001; Out 1995; Recombinant human FSH Study Group 1995; Schats 2000; Selman 2002; Westergaard 2001; Ye 2012);
- three used a short GnRH agonist protocol (Cheon 2004; Hugues 2001; Rashidi 2005); and

- two used no down-regulation protocol (Jansen 1998; Meden-Vrtovec 2003).

There is probably little to no difference in live birth between rFSH versus urinary gonadotropins with:

- antagonist protocol (OR 0.86, 95% CI 0.70 to 1.05; $I^2 = 0\%$; 4 studies, 1729 participants);
- long agonist protocol (OR 0.97, 95% CI 0.86 to 1.08; $I^2 = 0\%$; 25 studies, 6808 participants,
- short agonist protocol (OR 0.84, 95% CI 0.54 to 1.31; $I^2 = 0\%$; 3 studies, 402 participants);
- no down-regulation protocol (OR 1.17, 95% CI 0.62 to 2.23; $I^2 = 0\%$; 2 studies, 220 participants).

The test for subgroup differences did not suggest heterogeneity between subgroups ($\chi^2 = 1.69$, $P = 0.64$; $I^2 = 0\%$).

Live birth (or ongoing pregnancy) by sponsor

Of the 34 studies:

- 20 studies were sponsored by pharmaceutical companies:
 - Serono (6 studies; Bergh 1997; Franco 2000; Frydman 2000; Lenton 2000; Recombinant human FSH Study Group 1995; Schats 2000);
 - Organon (3 studies; Hedon 1995; Jansen 1998; Out 1995);
 - Ferring (9 studies; Andersen 2006; Devroey 2012; Dickey 2002; Dickey 2003; European and Israeli Study Group 2002; Hompes 2008; NCT00257556; Westergaard 2001; Witz 2020);
 - IBSA (2 studies; Aboulghar 2010; Baker 2009).
- 11 studies had no pharmaceutical funding (Balasch 2003; Bosch 2008; Cheon 2004; Gordon 2001; Hugues 2001; Kilani 2003; Meden-Vrtovec 2003; Mohamed 2006; Ng 2001; Rashidi 2005; Selman 2002);
- three studies' funding resources were unknown (Nardo 2000; Sohravand 2012; Ye 2012).

In studies sponsored by Ferring, rFSH may decrease live birth compared to urinary gonadotropins (OR 0.84, 95% CI 0.73 to 0.96; $I^2 = 0\%$; 9 studies, 4266 participants). This was expected as these nine studies all compared rFSH with HMG/HP-HMG.

In the other sponsor groups, there is probably little to no difference in live birth:

- Serono (OR 1.25, 95% CI 0.97 to 1.61; $I^2 = 0\%$; 6 studies, 1410 participants);
- Organon (OR 1.29, 95% CI 0.96 to 1.73; $I^2 = 0\%$; 3 studies, 1215 participants);
- IBSA (OR 1.04, 95% CI 0.62 to 1.74; $I^2 = 0\%$; 2 studies, 236 participants);
- no pharmaceutical funding OR 0.87, 95% CI 0.70 to 1.09; $I^2 = 0\%$; 11 studies, 1635 participants;
- sponsor unknown (OR 0.77, 95% CI 0.48 to 1.23; $I^2 = 10\%$; 3 studies, 397 participants).

The test for subgroup differences showed substantial heterogeneity between subgroups ($\chi^2 = 13.35$, $P = 0.02$; $I^2 = 62.5\%$).

OHSS by down-regulation protocol

Of the 37 studies that reported on OHSS:

- four used an antagonist protocol (Bosch 2008; Devroey 2012; NCT00257556; Witz 2020)
- 29 used a long GnRH agonist protocol (Aboulghar 2010; Andersen 2006; Antoine 2007; Baker 2009; Balasch 2003; Bergh 1997; Dickey 2002; Dickey 2003; European and Israeli Study Group 2002; Franco 2000; Frydman 2000; Gallego 2003; Gordon 2001; Hedon 1995; Hompes 2008; Hoomans 1999; Kilani 2003; Lenton 2000; Liu 2015; Mohamed 2006; Nardo 2000; Ng 2001; Out 1995; Recombinant human FSH Study Group 1995; Schats 2000; Selman 2002; Sohrabvand 2012; Westergaard 2001; Ye 2012)
- two used a short GnRH agonist protocol (Hugues 2001; Rashidi 2005); and
- two used no down-regulation protocol (Jansen 1998; Meden-Vrtovec 2003).

Results suggested that OHSS may be increased in rFSH compared to urinary gonadotropins when using a GnRH antagonist protocol (OR 2.00, 95% CI 1.37 to 2.93; $I^2 = 23\%$; 4 studies, 1729 participants), however, it should be noted that in all four studies, the urinary gonadotropin was HP-HMG.

There is probably little to no difference in OHSS between rFSH versus urinary gonadotropins when using a long GnRH agonist protocol (OR 1.14, 95% CI 0.85 to 1.54; $I^2 = 0\%$; 29 studies, 7746 participants).

There was no incidence of OHSS in the subgroups using a short GnRH agonist protocol or no down-regulation.

The test for subgroup differences showed substantial heterogeneity between subgroups ($\chi^2 = 5.14$, $P = 0.02$; $I^2 = 80.5\%$).

OHSS by sponsor

Of the 37 studies:

- 22 were sponsored by pharmaceutical companies:
 - Serono (6 studies; Bergh 1997; Franco 2000; Frydman 2000; Lenton 2000; Recombinant human FSH Study Group 1995; Schats 2000);
 - Organon (4 studies; Hedon 1995; Hoomans 1999; Jansen 1998; Out 1995);
 - Ferring (9 studies; Andersen 2006; Devroey 2012; Dickey 2002; Dickey 2003; European and Israeli Study Group 2002; Hompes 2008; NCT00257556; Westergaard 2001; Witz 2020);
 - IBSA (3 studies; Aboulghar 2010; Antoine 2007; Baker 2009)
- 11 studies had no pharmaceutical funding (Balasch 2003; Bosch 2008; Gordon 2001; Hugues 2001; Kilani 2003; Liu 2015; Meden-

Vrtovec 2003; Mohamed 2006; Ng 2001; Rashidi 2005; Selman 2002); and

- four studies' funding resources were unknown (Gallego 2003; Nardo 2000; Sohrabvand 2012; Ye 2012).

Results suggested OHSS increased in rFSH compared to urinary gonadotropins in studies sponsored by Serono (OR 2.16, 95% CI 1.10 to 4.25; $I^2 = 0\%$; 6 studies, 1420 participants) and Ferring (OR 1.58, 95% CI 1.16 to 2.16; $I^2 = 41\%$; 9 studies, 4266 participants).

OHSS in studies sponsored by Organon was also increased in rFSH but with a wide 95% CI that included no effect (OR 1.58, 95% CI 0.75 to 3.31; $I^2 = 0\%$; 4 studies, 1384 participants).

With only two OHSS cases, the evidence is uncertain for studies by IBSA (OR 0.99, 95% CI 0.14 to 7.16; $I^2 = 0\%$; 3 studies, 387 participants).

There is probably little to no difference in OHSS between rFSH and urinary gonadotropins in studies without pharmaceutical funding (OR 0.62, 95% CI 0.28 to 1.37; $I^2 = 0\%$; 11 studies, 1889 participants) or in studies where the sponsor was unknown (OR 0.86, 95% CI 0.39 to 1.88; $I^2 = 13\%$; 4 studies, 497 participants).

The test for subgroup differences showed no evidence of important subgroup differences ($\chi^2 = 7.90$, $P = 0.16$; $I^2 = 36.7\%$).

As no studies used a freeze-all policy, no additional analyses comparing fresh versus frozen were performed.

5. rFSH versus follitropin delta (recombinant hFSH)

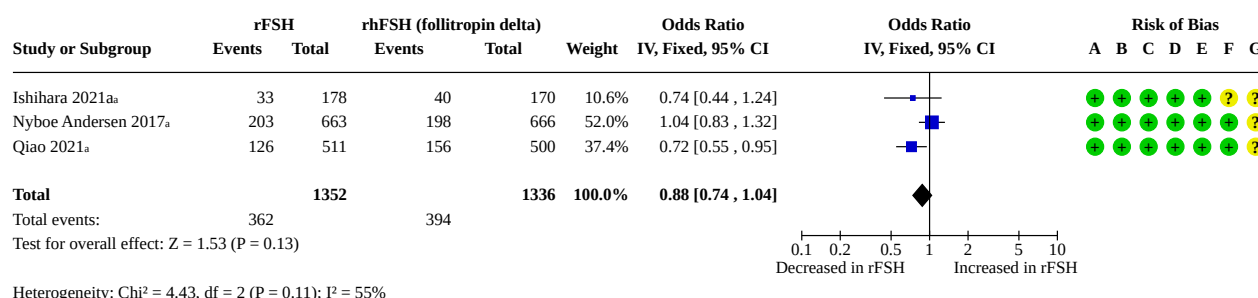
Data are presented in [Supplementary material 6](#) under 'Analysis group 5: rFSH versus recombinant hFSH (follitropin delta)' and [Summary of findings 3](#).

We included four studies for this comparison (EUCTR2021-001785-38-ES; Ishihara 2021a; Nyboe Andersen 2017; Qiao 2021).

Critical outcomes

Live birth (or ongoing pregnancy)

There is probably little or no difference between rFSH and follitropin delta (OR 0.88, 95% CI 0.74 to 1.04; $I^2 = 55\%$; 3 studies, 2688 participants; [Figure 9](#), moderate-certainty evidence). If the chance of achieving live birth with follitropin delta is 29%, the chance of live birth with rFSH is between 24% and 30%. Given the I^2 statistic of 55%, we carried out a sensitivity analysis using a random-effects model, which resulted in a comparable OR of 0.85 (95% CI 0.65 to 1.11).

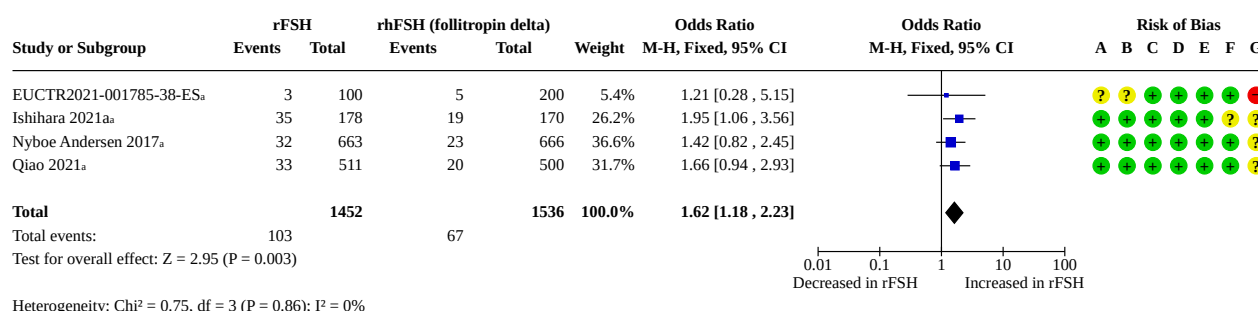
Figure 9. Comparison rFSH versus recombinant hFSH (follitropin delta) – outcome live birth (or ongoing pregnancy)**Footnotes**^aall studies were sponsored by Ferring, used antagonist protocol and fresh cycle**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

OHSS

OHSS is probably higher with rFSH compared to follitropin delta (OR 1.62, 95% CI 1.18 to 2.23; I² = 0%; 4 studies, 2988 participants;

Figure 10, moderate-certainty evidence). If the risk of OHSS with follitropin delta is 4.4%, the risk of OHSS with rFSH is between 5.1% and 9.2%.

Figure 10. Comparison rFSH versus recombinant hFSH (follitropin delta) – outcome OHSS**Footnotes**^aall studies were sponsored by Ferring, used antagonist protocol and fresh cycle**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

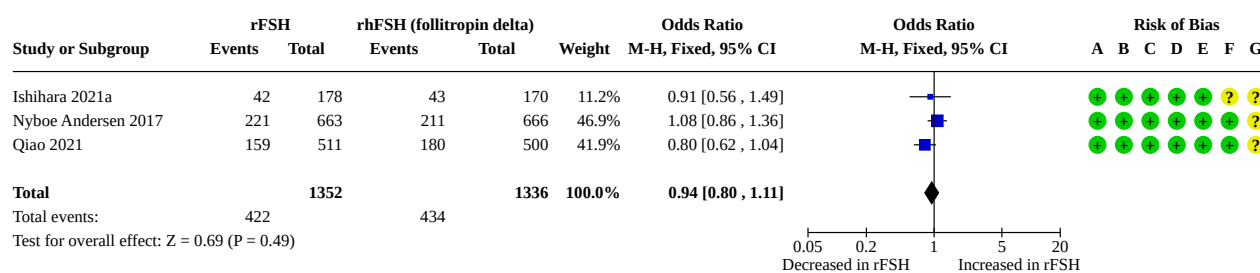
All studies were sponsored by Ferring, and used antagonist protocol and fresh cycle. Therefore, we did not perform any further subgroup analyses.

Important outcomes**Cumulative live birth (or ongoing pregnancy)**

No studies in this comparison reported this outcome.

Clinical pregnancy

There is no difference in clinical pregnancy when comparing rFSH to follitropin delta (OR 0.94, 95% CI 0.80 to 1.11; I² = 28%; 3 studies, 2688 participants; Figure 11, high-certainty evidence). If the chance of achieving a clinical pregnancy with follitropin delta is 32%, then the chance of a clinical pregnancy with rFSH is between 28% and 35%.

Figure 11. Comparison rFSH versus recombinant hFSH (follitropin delta) - outcome clinical pregnancy**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Multiple pregnancy

We are uncertain whether there may be a difference in multiple pregnancy when comparing rFSH to follitropin delta (OR 2.02, 95% CI 0.61 to 6.75; 1 study, 1329 participants).

Miscarriage

There may be little or no difference in miscarriage rate when comparing rFSH to follitropin delta (OR 1.66, 95% CI 0.78 to 3.55; 1 study, 1329 participants).

Ectopic pregnancy

No studies in this comparison reported this outcome.

Stillbirth

No studies in this comparison reported this outcome.

Termination of pregnancy

No studies in this comparison reported this outcome.

Gestational age at birth

There may be little or no difference in gestational age at birth when comparing rFSH to follitropin delta (MD 0.00, 95% CI -3.15 to 3.15; 1 study, 409 participants).

Birthweight

There may be little or no difference in birthweight when comparing rFSH to follitropin delta (MD -18.00, 95% CI -136.72 to 100.72; 1 study, 409 participants).

Neonatal mortality

We are uncertain whether there may be a difference in neonatal mortality when comparing rFSH to follitropin delta (OR 5.04, 95% CI 0.24 to 105.13; 3 studies, 2688 participants).

Major congenital anomalies

No studies in this comparison reported this outcome.

Amount of gonadotropins

Due to high heterogeneity, we did not pool the results in meta-analysis (Analysis 5.9).

Duration of ovarian stimulation

There may be a difference in the duration of ovarian stimulation when comparing rFSH to follitropin delta (MD -0.31, 95% CI -0.44 to -0.18; I² = 57%; 4 studies, 2982 participants).

Number of oocytes produced

Due to high heterogeneity, we did not pool the results in meta-analysis (Analysis 5.11).

Three studies reported on different dosages of follitropin delta (human cell line). We did not include them in the meta analyses (Arce 2014; Griesinger 2020; Ishihara 2021b).

Number of embryos (as reported by authors)

Three studies reported on the number of embryos (EUCTR2021-001785-38-ES; Nyboe Andersen 2017; Qiao 2021). Results are presented in Table 2.

Number of embryos frozen

One study reported on the number of embryos frozen (Nyboe Andersen 2017). Results are presented in Table 3.

All included studies used an antagonist protocol, fresh cycle and were sponsored by Ferring, thus we did not perform any subgroups.

Supporting analysis**Sensitivity analyses**

We carried out sensitivity analyses restricted to studies that we judged to be at low risk of selection bias.

Live birth (or ongoing pregnancy)

This comparison included three studies in the sensitivity analysis (Ishihara 2021a; Nyboe Andersen 2017; Qiao 2021), which we also included in the main analysis (OR 0.88, 95% CI 0.74 to 1.04; I² = 55%; 3 studies, 2688 participants).

OHSS

This comparison included three studies in the sensitivity analysis (Ishihara 2021a; Nyboe Andersen 2017; Qiao 2021). OHSS is probably increased in rFSH compared to follitropin delta (OR 1.64, 95% CI 1.18 to 2.29; $I^2 = 0\%$; 3 studies, 2688 participants).

As no studies used a freeze-all policy, we did not perform any additional analyses comparing fresh versus frozen.

6. rFSH versus biosimilars

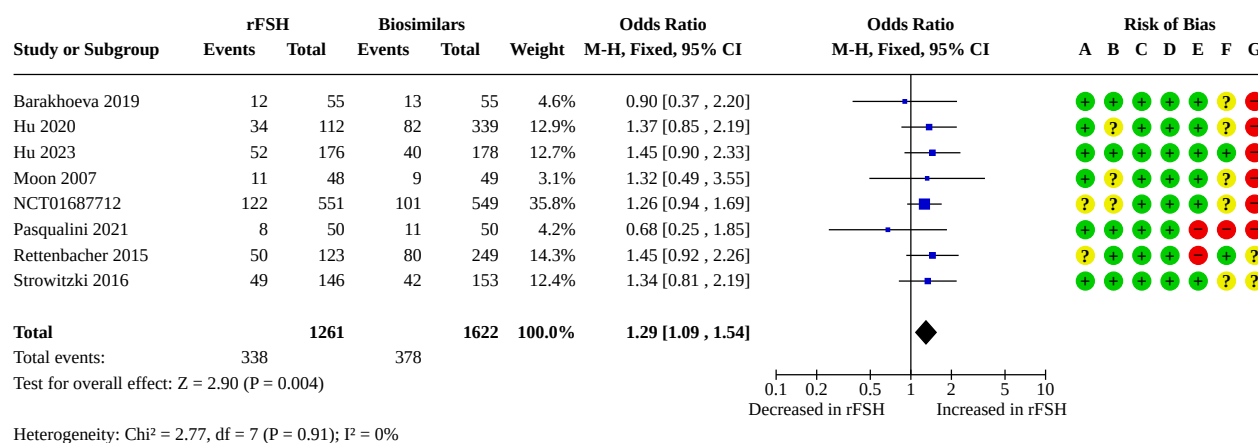
Data are presented in [Supplementary material 6](#) under 'Analysis group 6: rFSH versus biosimilars' and [Summary of findings 4](#).

The main analysis for this comparison included eight studies (Barakhoeva 2019; Hu 2020; Hu 2023; Moon 2007; NCT01687712; Pasqualini 2021; Rettenbacher 2015; Strowitzki 2016).

Critical outcomes**Live birth (or ongoing pregnancy)**

Live birth is probably increased in rFSH compared to biosimilars (OR 1.29, 95% CI 1.09 to 1.54; $I^2 = 0\%$; 8 studies, 2883 participants; [Figure 12](#), moderate-certainty evidence). If the chance of achieving live birth with biosimilars is 23%, the chance of live birth with rFSH is between 25% and 32%.

Figure 12. Comparison rFSH versus biosimilars – outcome live birth (or ongoing pregnancy)

**Risk of bias legend**

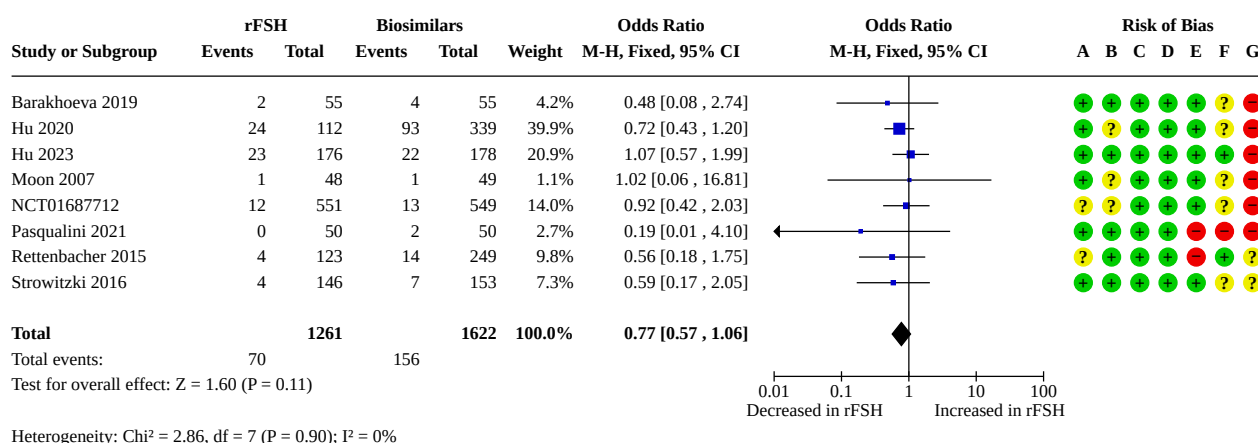
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

OHSS

There may be little to no difference in OHSS when comparing rFSH to biosimilars (OR 0.77, 95% CI 0.57 to 1.06; $I^2 = 0\%$; 8 studies, 2883

participants; [Figure 13](#), low-certainty evidence). If the risk of OHSS with biosimilars is 9.6%, the chance of OHSS with rFSH is between 5.7% and 10.1%.

Figure 13. Comparison rFSH versus biosimilars - outcome OHSS

**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

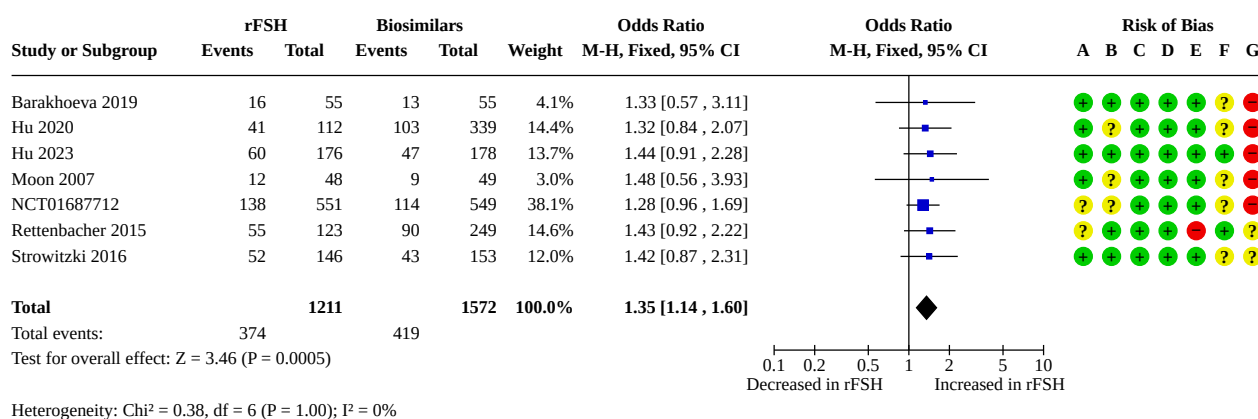
Important outcomes**Cumulative live birth (or cumulative ongoing pregnancy)**

One study reported on cumulative live birth for rFSH versus biosimilars (Strowitzki 2016). Although the results favoured rFSH, the confidence interval crossed the line of no effect (OR 1.30, 95% CI 0.79 to 2.14; 1 study, 299 participants).

Clinical pregnancy

Clinical pregnancy is probably increased in rFSH when compared to biosimilars (OR 1.35, 95% CI 1.14 to 1.60; I² = 0%; 7 studies, 2783 participants; Figure 14, moderate-certainty evidence). If the chance of a clinical pregnancy with biosimilars is 27%, the chance of live birth with rFSH is between 29% and 37%.

Figure 14. Comparison rFSH versus biosimilars - outcome clinical pregnancy

**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Multiple pregnancy

There may be little to no difference in multiple pregnancy when comparing rFSH versus biosimilars (OR 0.98, 95% CI 0.49 to 1.94; $I^2 = 0\%$; 4 studies, 606 participants).

Miscarriage

There may be little to no difference in miscarriage rate when comparing rFSH to biosimilars (OR 1.55, 95% CI 0.84 to 2.84; $I^2 = 0\%$; 4 studies, 1606 participants).

Ectopic pregnancy

There may be little to no difference in ectopic pregnancy when comparing rFSH to biosimilars (OR 0.86, 95% CI 0.29 to 2.58; $I^2 = 0\%$; 4 studies, 1609 participants).

Stillbirth

There may be little to no difference in stillbirth when comparing rFSH to biosimilars (OR 3.08, 95% CI 0.32 to 29.70; $I^2 = 0\%$; 2 studies, 1399 participants).

Termination of pregnancy

There may be little to no difference in termination of pregnancy when comparing rFSH to biosimilars (OR 0.65, 95% CI 0.17 to 2.49; $I^2 = 49\%$; 2 studies, 1399 participants).

Gestational age at birth

No studies in this comparison reported this outcome.

Birthweight

No studies in this comparison reported this outcome.

Neonatal mortality

There may be little to no difference in neonatal mortality when comparing rFSH to biosimilars (OR 1.00, 95% CI 0.06 to 15.97; 1 study, 1100 participants).

Major congenital anomalies

There may be little to no difference in major congenital anomalies when comparing rFSH to biosimilars (OR 1.30, 95% CI 0.57 to 3.00; 1 study, 1100 participants).

Patient acceptability/satisfaction

There may be little to no difference in patient acceptability/satisfaction when comparing rFSH to biosimilars (OR 0.47, 95% CI 0.08 to 2.61; 1 study, 299 participants).

Amount of gonadotrophins used

There may be little to no difference in the amount of gonadotrophins used when comparing rFSH to biosimilars (MD 12.29, 95% CI -17.05 to 41.62; $I^2 = 53\%$; 8 studies, 2851 participants).

Duration of ovarian stimulation

Due to high heterogeneity, we did not pool the results in meta-analysis (Analysis 6.14).

Number of oocytes produced per cycle

There may be little or no difference in the number of oocytes produced per cycle when comparing rFSH to biosimilars (MD -1.94, 95% CI -2.11 to -1.77; $I^2 = 81\%$; 8 studies, 2795 participants).

Number of embryos (as reported by study authors)

Available results are presented in [Table 2](#).

Number of embryos frozen

Available results are presented in [Table 3](#).

Supporting analyses

Live birth (or ongoing pregnancy) by down-regulation

Of the eight studies included:

- three studies used a GnRH antagonist protocol (Barakhoeva 2019; Moon 2007; Pasqualini 2021); and
- five studies used a long GnRH agonist protocol (Hu 2020; Hu 2023; NCT01687712; Rettenbacher 2015; Strowitzki 2016).

There is probably little to no difference in live birth when comparing rFSH to biosimilars in antagonist cycles (OR 0.93, 95% CI 0.54 to 1.62; $I^2 = 0\%$; 3 studies, 307 participants). However, results showed an increase in live birth when using rFSH versus biosimilars in long GnRH agonist cycles (OR 1.34, 95% CI 1.12 to 1.62; $I^2 = 0\%$; 5 studies, 2576 participants).

The test for subgroup differences did not show evidence of substantial heterogeneity between subgroups ($\chi^2 = 1.53$, $P = 0.22$; $I^2 = 34.6\%$).

Live birth (or ongoing pregnancy) by sponsor

Of the eight studies included:

- seven were sponsored by pharmaceutical companies (Barakhoeva 2019; Hu 2020; Hu 2023; Moon 2007; NCT01687712; Rettenbacher 2015; Strowitzki 2016); and
- one study's sponsorship was unknown (Pasqualini 2021).

Live birth is probably increased in rFSH compared to biosimilars in studies sponsored by pharmaceutical companies (OR 1.32, 95% CI 1.11 to 1.58; $I^2 = 0\%$; 7 studies, 2783 participants, moderate-certainty evidence). There may be no difference in live birth in the study whose sponsorship was unknown (OR 0.68, 95% CI 0.25 to 1.85; 1 study, 100 participants, low-certainty evidence).

The test for subgroup differences did not show evidence of substantial heterogeneity between subgroups ($\chi^2 = 1.65$, $P = 0.20$; $I^2 = 39.4\%$).

OHSS by down-regulation

Of the eight studies included in the main analysis:

- three studies used a GnRH antagonist protocol (Barakhoeva 2019; Moon 2007; Pasqualini 2021); and
- five studies used a long GnRH agonist protocol (Hu 2020; Hu 2023; NCT01687712; Rettenbacher 2015; Strowitzki 2016).

There may be little to no difference in OHSS when comparing rFSH versus biosimilars in antagonist cycles (OR 0.45, 95% CI 0.13 to 1.65; $I^2 = 0\%$; 3 studies, 307 participants), or in long agonist cycles (OR 0.80, 95% CI 0.58 to 1.11; $I^2 = 0\%$; 5 studies, 2576 participants).

The test for subgroup differences did not show evidence of heterogeneity between subgroups ($\chi^2 = 0.70$, $P = 0.40$; $I^2 = 0\%$).

OHSS by sponsor

- Seven studies were sponsored by pharmaceutical companies (Barakhoeva 2019; Hu 2020; Hu 2023; Moon 2007; NCT01687712; Rettenbacher 2015; Strowitzki 2016).
- The sponsorship of one study was unknown (Pasqualini 2021).

There is probably little to no difference in OHSS when comparing rFSH versus biosimilars in studies sponsored by pharmaceutical companies (OR 0.79, 95% CI 0.58 to 1.08; $I^2 = 0\%$; 7 studies, 2783 participants). It is uncertain whether there is a difference in OHSS in the single study with unknown sponsorship (OR 0.19, 95% CI 0.01 to 4.10; 1 study, 100 participants).

The test for subgroup differences did not show evidence of heterogeneity between subgroups ($\text{Chi}^2 = 0.81$, $P = 0.37$; $I^2 = 0\%$).

As no studies used a freeze-all policy, we did not perform any additional analyses comparing fresh versus frozen.

Sensitivity analyses

Live birth (or ongoing pregnancy)

We carried out sensitivity analyses restricted to four studies at low risk of selection bias (Barakhoeva 2019; Hu 2023; Pasqualini 2021; Strowitzki 2016). Results suggested that there is probably little or no difference in live birth when comparing rFSH versus biosimilars (OR 1.24, 95% CI 0.91 to 1.68; $I^2 = 0\%$; 4 studies, 863 participants).

OHSS

We carried out sensitivity analyses restricted to four studies at low risk of selection bias (Barakhoeva 2019; Hu 2023; Pasqualini 2021; Strowitzki 2016). Results suggested that there may be little or no difference in OHSS when comparing rFSH to biosimilars (OR 0.83, 95% CI 0.50 to 1.39; $I^2 = 0\%$; 4 studies, 863 participants).

Additional sensitivity analyses

We performed sensitivity analyses for all critical outcomes by using a random-effects model, instead of a fixed-effect model. There is no difference in results for all outcomes (data not shown).

Equity assessment

The studies included in this review were conducted in North America, Europe, Asia and Africa, and involved populations of diverse ethnic backgrounds. Although existing literature suggests that certain groups — such as those with higher BMI or suboptimal lifestyle factors — may have lower chances of success, these factors are unlikely to have influenced the relative effectiveness of the interventions, given the randomised design of the included studies.

With regard to outcomes, pregnancy-related endpoints are objective and universally applicable. Therefore, we consider the findings to be generalisable to all women with an indication for IVF or ICSI, regardless of background.

Reporting biases

We reported our assessment of the risk of selective outcome reporting for each included study in [Risk of bias in included studies](#). We also planned to create funnel plots to investigate the possibility of publication bias for the primary analyses (critical outcomes and clinical pregnancy). Only the comparisons rFSH versus HMG/HP-HMG and rFSH vs FSH-HP contained more than 10 studies. The

funnel plots did not indicate that there was publication bias (data not shown).

DISCUSSION

Summary of main results

This review compared the effectiveness of recombinant FSH (rFSH) with HMG/HP-HMG ([Summary of findings 1](#)), FSH-HP ([Summary of findings 2](#)), follitropin delta ([Summary of findings 3](#)), and biosimilars ([Summary of findings 4](#)), in women undergoing ovarian stimulation for IVF or ICSI.

Overall, rFSH probably resulted in a slightly lower live birth and clinical pregnancy when compared with HMG/HMG-HP, while there was little to no difference in live birth and clinical pregnancies when rFSH was compared with FSH-HP. OHSS was probably higher with rFSH when compared to HMG/HMG-HP, while there may be little to no difference in OHSS for rFSH versus FSH-HP. When we included only studies at low risk of selection bias, the difference in live birth between rFSH and HMG/HMG-HP may no longer be present, while the other analyses remained similar.

Between rFSH and follitropin delta, there was probably little to no difference in live birth, and high-certainty evidence of no difference in clinical pregnancy. However, OHSS was probably increased with rFSH. Sensitivity analyses including only studies at low risk of selection bias yielded similar results.

When compared with biosimilars, rFSH probably increased live birth and clinical pregnancies, while there may be little to no difference in OHSS risk. However, the sensitivity analysis, including only studies at low risk of selection bias, showed that there may be no difference in live birth when comparing rFSH with biosimilars.

Limitations of the evidence included in the review

Most of the included studies used computer-generated randomisation. However, the method of allocation concealment was not always specified, leading to an unclear risk of bias in some studies. The certainty of evidence was generally moderate, although downgrading was occasionally done for imprecision, or inconsistency. The authors of this review had some concerns about the trustworthiness of many of the included studies, which also resulted in a downgrading of the level of evidence. These concerns were usually related to an insufficient description of allocation concealment, zero losses to follow-up reported or perfectly rounded numbers of participants in each study group.

The majority of included studies reported on the critical outcomes live birth and OHSS. Data from studies comparing rFSH versus FSH-P and FSH-HP were more incomplete as these were often published at a time when clinical pregnancy was still an accepted endpoint. Live birth data were primarily reported after fresh embryo transfer, with only a few studies providing cumulative live birth outcomes. The observed lower live birth with rFSH versus HMG/HP-HMG, and the higher rate with rFSH versus biosimilars, may not persist when cumulative live births, including frozen-thawed cycles, are considered. Additionally, the increasing use of freeze-all cycles may eliminate the observed differences in OHSS risk.

The sensitivity analyses, which included only studies at low risk of selection bias, showed different results for live birth in both the comparison of rFSH versus HMG/HMG-HP and biosimilars.

Studies that were excluded from the sensitivity analysis mainly lacked detailed information on randomisation and allocation concealment, including no reporting of the opaqueness of the used envelopes or the absence of any clarification of the used methods for allocation concealment or randomisation.

The data on the amount of gonadotropins used, the number of oocytes retrieved, and the duration of stimulation were generally reported not per woman randomised, but rather per woman who underwent ovarian stimulation or oocyte retrieval. Moreover, due to high heterogeneity across studies in most comparisons, these outcomes were usually not pooled in meta-analyses.

Most studies did not evaluate the preference of patients for a particular type of gonadotropin. However, as all gonadotropins are given subcutaneously with very few clinical differences, we would not expect to see any clinically important impact of such preference.

The majority of the included studies were industry-sponsored, which may have introduced bias favouring the gonadotropin produced by the sponsor. Subgroup analyses grouping primary outcomes by pharmaceutical sponsorship suggested that live birth and clinical pregnancies were lower for rFSH compared to urinary FSH in the Ferring-sponsored trials, all of which involved HMG/HP-HMG. However, further analysis comparing Ferring-sponsored and non-sponsored trials that assessed HMG/HP-HMG versus rFSH showed similar summary odds ratios with overlapping confidence intervals. Therefore, although a sponsor effect cannot be entirely ruled out, any potential bias appears to be limited.

The evidence is broadly applicable to standard IVF and ICSI cycles. All studies involved fresh embryo transfer, with only six also reporting subsequent frozen-thawed cycles. Consequently, these results may not fully capture outcomes from freeze-all cycles.

Limitations of the review processes

We performed subgroup analyses only for the different methods of down-regulation used and for studies with pharmaceutical sponsoring. Any other differences in study protocols, such as the fertilisation method used, the starting dose of gonadotropins or the number of embryos transferred, were not further explored in the current review.

Agreements and disagreements with other studies or reviews

Urinary gonadotropins

Our results agree with most of the previous reviews that compared rFSH versus urinary gonadotropins (196, 197, 198, 199, 200, 201, 202, 203). Three older reviews suggested a higher live birth with either FSH-HP or HMG compared to rFSH (204, 199, 201), whereas one older review found a higher clinical pregnancy in rFSH versus FSH-HP/FSH-P [197]. These discrepancies may be explained by differences in newly included studies in the more recent reviews. Our findings are in accordance with the latest European Society of Human Reproduction and Embryology (ESHRE) guideline on ovarian stimulation for IVF and ICSI [205].

Follitropin delta

Several other reviews compared rFSH versus follitropin delta but reported different results compared to the current review. One

review described an increased live birth and a decreased incidence of OHSS in favour of follitropin delta in women with higher AMH levels (206). In contrast, the other reviews reported a similar live birth in rFSH versus follitropin delta (207, 208, 209). The risk of OHSS was found to be similar in rFSH versus follitropin delta in one review (208), but was increased in another review (209). One review did not estimate the risk of OHSS in their meta-analysis (207). Importantly, there were differences in the included studies between these reviews and the current review; for example, we included a relatively recent study (EUCTR2021-001785-38-ES), whereas other reviews included dose-response trials (Arce 2014; Ishihara 2021b) or pharmacokinetics studies.

Biosimilars

Our results agree with one other systematic review that compared rFSH versus biosimilars and described an increased chance of achieving a live birth when using rFSH (210).

AUTHORS' CONCLUSIONS

Implications for practice

Compared to human menopausal gonadotropin (HMG) or highly purified HMG (HP-HMG), recombinant follicle stimulating hormone (rFSH) showed slightly lower live birth and clinical pregnancies while increasing the risk of ovarian hyperstimulation syndrome (OHSS). When rFSH was compared to purified urinary follicle-stimulating hormone (FSH-P) or highly purified urinary FSH (FSH-HP), there were probably no differences in live birth and clinical pregnancies nor in OHSS risk. When compared with follitropin delta (recombinant hFSH), there was probably little or no difference in live birth or clinical pregnancies, while OHSS was probably increased with rFSH. When compared with biosimilars, live birth and clinical pregnancies were probably higher with rFSH, while there may be little or no difference in OHSS. Besides effectiveness and safety, the choice of one or the other product will depend upon the availability of the product, the convenience of its use, and the associated costs.

Implications for research

We included 59 studies involving 18,119 women. For the main outcomes, the certainty of the evidence is moderate, suggesting that additional studies addressing the same primary comparisons are not urgently needed. Future research should instead focus on cumulative outcomes across multiple frozen embryo transfer cycles, as these provide a more complete picture of treatment effectiveness. In addition, strategies for the prevention and management of OHSS should remain a priority, given its clinical importance and impact on patient safety.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD005354.pub3](https://doi.org/10.1002/14651858.CD005354.pub3).

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of studies awaiting classification

Supplementary material 5 Characteristics of ongoing studies

Supplementary material 6 Analyses

Supplementary material 7 Data package

Supplementary material 8 Differences between protocol and review

Supplementary material 9 Criteria for judging risk of bias with the RoB 1 assessment tool

Supplementary material 10 Summary of findings table – rFSH compared to FSH-P for ovarian stimulation in assisted reproductive technology cycles

ADDITIONAL INFORMATION

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Editorial and peer-reviewer contributions

Cochrane Gynaecology and Fertility supported the authors in the development of this review update.

The following people conducted the editorial process for this article:

- Sign-off Editor (final editorial decision): Vanessa Jordan, Faculty of Medical and Health Sciences, University of Auckland;
- Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Jenny Bellorini, Cochrane Central Editorial Service;
- Editorial Assistant (conducted editorial policy checks, selected peer reviewers, collated peer-reviewer comments and supported the editorial team): Cynthia Stafford, Cochrane Central Editorial Service;
- Copy Editor (copy editing and production): Denise Mitchell, Cochrane Central Production Service;
- Peer-reviewers (provided comments and recommended an editorial decision): Roger Hart MBBS FRANZCOG FRCOG MD CREI MBA, Professor of Reproductive Medicine, Medical Director of Fertility Specialists of WA and National Medical Director of City Fertility, Division of Obstetrics and Gynaecology, The University of Western Australia (clinical/content review); Dr Tejaswini Arunachala Murthy PhD (acute care medicine - intensive care nutrition), MBBS, DA, IDCCM, NSP, PDCR, DipECHO, Invited grant assessor, NHMRC, Clinical Lecturer University of Adelaide (patient and public review); Nuala Livingstone, Cochrane Evidence Production and Methods Directorate (methods review); Jo Platt, Central Editorial Information Specialist (search review).

Contributions of authors

RPB searched the literature, performed the risk of bias and TRACT assessments, entered the data, did the analyses, wrote the review.

What's new

EBK searched the literature, performed the risk of bias and TRACT assessments, entered the data, did the analyses, wrote the review.

MvW searched the literature, contacted authors of studies, entered the data, did the analyses, wrote the review, and responded to reviewer comments in the previous version of this review.

Some authors involved in the previous published version of this review are no longer included on the author byline: Dr Irene Kwan, Dr Anna Burt, Dr Jane Thomas, Professor Andy Vail, Professor Fulco van de Veen and Professor Hesham Al-Inany. Some of the content retained in this review reflects their contributions.

Declarations of interest

RPB has no conflict of interest to declare.

EBK is a staff member of Cochrane Gynaecology and Fertility and has no other conflicts to declare. She was not involved in the editorial processing of the review update.

MvW is a staff member of Cochrane Gynaecology and Fertility and has no other conflicts to declare. She was not involved in the editorial processing of the review update.

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Internal sources

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External sources

- No sources of support provided

Registration and protocol

Protocol (2005) DOI: 10.1002/14651858.CD005354 [211]

Original review (2011) DOI: 10.1002/14651858.CD005354.pub2 [12]

Data, code and other materials

As part of the published Cochrane review, we have made the following available for download for users of the Cochrane Library (see data package [Supplementary material 7](#)): full search strategies for each database ([Supplementary material 1](#)); references, study characteristics, and risk of bias assessments of all included studies ([Supplementary material 2](#)); references and reasons for exclusion of all studies excluded at full stage screening ([Supplementary material 3](#)); references to studies awaiting classification ([Supplementary material 4](#)); all analyses performed in RevMan Web ([Supplementary material 6](#)).

Notes

Three previously published Cochrane reviews have been superseded by this review. These three reviews are now withdrawn [13, 14, 15].

Date	Event	Description
14 July 2026	New citation required and conclusions have changed	We added a comparison with biosimilars and recombinant FSH (human cell line). The conclusions of the review have changed with the addition of new studies (25 new studies added). The certainty of the evidence was low to moderate for the critical outcomes.
14 July 2026	New search has been performed	We updated the review.

History

Protocol first published: Issue 3, 2005

Review first published: Issue 2, 2011

Date	Event	Description
7 November 2012	Feedback has been incorporated	Abstract edited. Results for individual urinary gonadotropins included in abstract.
15 March 2011	Amended	Minor edit. Added to the abstract how many people selected studies and extracted data. Added dates to the searches in the abstract
15 March 2011	Amended	Minor edit. Added to the Plain language summary the word significantly: Comparing rFSH with HMG/HP-HMG resulted in a "significantly" lower live birth rate in the rFSH group though differences were small. Added the grading HIGH for the primary adverse outcome OHSS to the summary of findings table.
21 February 2011	Amended	Minor edit made EISG study was reported in the table as having been conducted with Puregon whereas it should be Gonal F- this is now corrected
3 February 2011	Amended	Minor edit to with the addition of the correct DOI to the reference Van Wely 2002
19 November 2007	New citation required and conclusions have changed	Substantive amendment of protocol

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ADDITIONAL TABLES

Recombinant follicle-stimulating hormone (rFSH) versus other recombinant or urinary gonadotropins for ovarian stimulation in assisted reproductive technology cycles (Review)

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Table 1. Overview of included studies and syntheses

Study name (year)	Population: sample size, age distribution	Intervention	Comparison	Critical outcomes
Abate 2009	n = 401, 26-38	rFSH (Gonal-F)	FSH-HP (Fostimon)	Live birth (or ongoing pregnancy)
Aboulghar 2010	n = 84, < 39	rFSH (Gonal F)	FSH-HP (Fostimon)	Live birth (or ongoing pregnancy), OHSS
Andersen 2006	n = 731, 21-37	rFSH (Gonal F)	HP-HMG (Menopur)	Live birth (or ongoing pregnancy), OHSS
Antoine 2007	n = 150, 18-39	rFSH (Gonal F)	FSH-HP (Fostimon)	OHSS
Arce 2014	n = 265, 18-37	rFSH (Gonal F)	rhFSH (FE 999049)	Live birth (or ongoing pregnancy), OHSS
Baker 2009	n = 152, 18-39	rFSH (Gonal-F)	FSH-HP (Fostimon)	Live birth (or ongoing pregnancy), OHSS
Balasch 2003	n = 60, 26-37	rFSH (Gonal-F)	HMG (Pergonal)	Live birth (or ongoing pregnancy), OHSS
Barakhoeva 2019	n = 110, 20-35	rFSH (Gonal-F)	Biosimilar (Primapur)	Live birth (or ongoing pregnancy), OHSS
Bergh 1997	n = 235, 18-38	rFSH (Gonal-F)	FSH-HP (Metrodin-HP)	Live birth (or ongoing pregnancy), OHSS
Bosch 2008	n = 280, 18-37	rFSH (Gonal-F)	HP-HMG (Menopur)	Live birth (or ongoing pregnancy), OHSS
Chapon 2021	n = 188	rFSH (Puregon)	HP-HMG (Menopur)	-
Cheon 2004	n = 241, 25-40	rFSH (Puregon)	FSH-HP (Metrodin-HP)	Live birth (or ongoing pregnancy)
Devroey 2012	n = 749, 21-34	rFSH (Puregon)	HP-HMG (Menopur)	Live birth (or ongoing pregnancy), OHSS
Dickey 2002	n = 177, 18-39	rFSH (Follistim)	FSH-HP (Bravelle)	Live birth (or ongoing pregnancy), OHSS
Dickey 2003	n = 120, 18-39	rFSH (Follistim)	FSH-HP (Bravelle)	Live birth (or ongoing pregnancy), OHSS
Drakakis 2002	n = 70	rFSH	FSH-HP	-
European and Israeli Study Group 2002	n = 781, 23-36	rFSH (Gonal-F)	HP-HMG (Menopur)	Live birth (or ongoing pregnancy), OHSS
EUC- TR2021-001785-38-ES	n = 300, 18-40	rFSH (Gonal-F)	Follitropin delta (Rekovele)	OHSS
Franco 2000	n = 120	rFSH (Gonal-F)	FSH-HP (Metrodin-HP)	Live birth (or ongoing pregnancy), OHSS
Frydman 2000	n = 278, 18-38	rFSH (Gonal-F)	FSH-HP (Metrodin-HP)	Live birth (or ongoing pregnancy), OHSS
Gallego 2003	n = 100, < 40	rFSH (Puregon)	FSH-HP (Neoferti- norm)	OHSS
Gholami 2010	n = 115, < 37	rFSH (Gonal F)	FSH-HP (Fostimon)	-
Gordon 2001	n = 128, 18-38	rFSH (Puregon)	HMG (Humegon)	Live birth (or ongoing pregnancy), OHSS
Griesinger 2020	n = 247, 18-37	rFSH (Gonal-F)	Follitropin epsilon	Live birth (or ongoing pregnancy), OHSS

Table 1. Overview of included studies and syntheses *(Continued)*

Hedon 1995	n = 99, 18-39	rFSH (Puregon)	FSH-P (Metrodin)	Live birth (or ongoing pregnancy), OHSS
Hompes 2008	n = 629, 18-39	rFSH (Gonal-F)	HP-HMG (Menopur)	Live birth (or ongoing pregnancy), OHSS
Hoomans 1999	n = 169, 18-39	rFSH (Puregon)	FSH-HP (Metrodin-HP)	OHSS
Hu 2020	n = 451, 20-39	rFSH (Gonal-F)	Biosimilar (Follitrope)	Live birth (or ongoing pregnancy); OHSS
Hu 2023	n = 354, 20-39	rFSH (Gonal-F)	Biosimilar (QL1012)	Live birth (or ongoing pregnancy), OHSS
Hugues 2001	n = 88, 18-38	rFSH (Gonal-F) or rFSH (Puregon)	FSH-HP (Metrodin-HP)	Live birth (or ongoing pregnancy), OHSS
Ishihara 2021a	n = 347, 20-40	rFSH (Follistim)	Follitropin delta (Rekovele)	Live birth (or ongoing pregnancy), OHSS
Ishihara 2021b	n = 158, 20-39	rFSH (Follistim)	Follitropin delta (Rekovele)	Live birth (or ongoing pregnancy), OHSS
Jansen 1998	n = 109, < 40	rFSH (Puregon)	HMG (Humegon)	Live birth (or ongoing pregnancy), OHSS
Kilani 2003	n = 100, 18-37	rFSH (Gonal-F)	HP-HMG (Menopur)	Live birth (or ongoing pregnancy), OHSS
Lenton 2000	n = 168, 18-38	rFSH (Gonal-F)	FSH-HP (Metrodin-HP)	Live birth (or ongoing pregnancy), OHSS
Liu 2015	n = 508, > 37	rFSH (Gonal F)	FSH-HP (Fostimon)	OHSS
Meden-Vrtovec 2003	n = 131	rFSH	FSH-P	-
Mohamed 2006	n = 257, > 39	rFSH (Gonal F)	FSH-HP (Fostimon)	-
Moon 2007	n = 97, 20-38	rFSH (Gonal-F)	Biosimilar (DA-3801)	Live birth (or ongoing pregnancy), OHSS
Nardo 2000	n = 110	rFSH (Gonal-F) or rFSH (Puregon)	FSH-HP (Metrodin-HP)	Live birth (or ongoing pregnancy), OHSS
NCT00257556	n = 80, 20-35	rFSH (Gonal-F)	HP-HMG (Menopur)	Live birth (or ongoing pregnancy), OHSS
NCT01687712	n = 1100, 35-42	rFSH (Gonal-F)	Biosimilar (AFOLIA)	Live birth (or ongoing pregnancy), OHSS
Ng 2001	n = 40, < 39	rFSH (Gonal-F)	HMG (Pergonal)	Live birth (or ongoing pregnancy), OHSS
Nyboe Andersen 2017	n = 1329, 18-40	rFSH (Gonal-F)	Follitropin delta (FE 999049)	Live birth (or ongoing pregnancy), OHSS
Out 1995	n = 1027, 18-39	rFSH (Puregon)	FSH-P (Metrodin)	Live birth (or ongoing pregnancy), OHSS
Pasqualini 2021	n = 100, 18-38	rFSH (Gonal-F)	Biosimilar (Folitime)	Live birth (or ongoing pregnancy), OHSS
Qiao 2021	n = 1011, 20-40	rFSH (Gonal-F)	Follitropin delta (Rekovele)	Live birth (or ongoing pregnancy), OHSS
Rashidi 2005	n = 60, < 35	rFSH (Gonal-F)	HMG	Live birth (or ongoing pregnancy), OHSS

Table 1. Overview of included studies and syntheses (Continued)

Rettenbacher 2015	n = 372, 20-38	rFSH (Gonal-F)	Biosimilar (Bemfola)	Live birth (or ongoing pregnancy), OHSS
Recombinant human FSH Study Group 1995	n = 127, 18-38	rFSH	FSH-P	Live birth (or ongoing pregnancy), OHSS
Schats 2000	n = 496, 18-38	rFSH (Gonal-F)	FSH-HP (Metrodin-HP)	Live birth (or ongoing pregnancy), OHSS
Selman 2002	n = 267, 18-38	rFSH (Gonal-F)	FSH-HP (Fostimon)	Live birth (or ongoing pregnancy), OHSS
Selman 2010	n = 188, 27-38	rFSH (Gonal-F)	FSH-HP (Fostimon)	OHSS
Sohrabvand 2012	n = 160, 20-35	rFSH (Gonal-F)	FSH-HP (Fostimon)	Live birth (or ongoing pregnancy), OHSS
Strehler 2001	n = 578, < 40	rFSH (Gonal-F)	HMG (Menogon)	-
Strowitzki 2016	n = 299, 18-37	rFSH (Gonal-F)	Biosimilar (Ovaleap)	Live birth (or ongoing pregnancy), OHSS
Westergaard 2001	n = 379, 21-39	rFSH (Gonal-F)	HMG (Menogon)	Live birth (or ongoing pregnancy), OHSS
Witz 2020	n = 620, 21-35	rFSH (Gonal-F)	HP-HMG (Menopur)	Live birth (or ongoing pregnancy), OHSS
Ye 2012	n = 127, 35-39	rFSH (Gonal-F)	HP-HMG (Menopur)	Live birth (or ongoing pregnancy), OHSS

FSH-HP: highly purified follicle-stimulating hormone; **FSH-P:** purified follicle-stimulating hormone; **HMG:** human menopausal gonadotropin; **HP-HMG:** highly purified human menopausal gonadotropin; **OHSS:** ovarian hyperstimulation syndrome; **rFSH:** recombinant follicle-stimulating hormone; **rhFSH:** recombinant human follicle-stimulating hormone

Table 2. Number of embryos

Study	Recombinant FSH			Urinary gonadotropins, rhFSH (human cell line), biosimilar		
	Mean	SD	N	Mean	SD	N
Andersen 2006	7.4	5	368	6.3	4.7	363
Baker 2009	9.6	6.1	76	9.2	5.8	76
Balasch 2003	7.95	0.65	25	4.86	0.55	25
Bergh 1997	8.1	4.2	119	4.7	3.5	102
Chapon 2021	4.17	3.1	85	3.26	2.4	83
Cheon 2004	1319		127	1304		121
Devroey 2012	4.8	3.7	375	4	3	374
EUCTR2021-001785-38-ES	2.83	2.63	100	2.7	2.99	200
Franco 2000	6.3	4.5	60	5.5	3.7	60

Table 2. Number of embryos *(Continued)*

Frydman 2000	5.0	3.7	128	3.5	2.9	113
Gallego 2003	5.88	3.98	43	5.42	4.4	45
Hedon 1995	3.7			4.0		
Hoomans 1999	6.4		83	5.4		82
Hugues 2001	3.37	2.9	48	3.4	2.9	30
Jansen 1998	4.1		47	4.4		32
Lenton 2000	5.5	3.6	80	5.5	2.8	75
Meden-Vrtovec 2003	2.8	2.2	70	2.3	1.9	61
Ng 2001	261		20	254		20
Nyboe Andersen 2017	5.7	4.3	643	5.4	3.7	635
Out 1995	3.11			2.61		
Pasqualini 2021	0.22	0.7	41	0.38	0.7	45
Qiao 2021	8.7	5.5	510	7.0	4.6	499
Rashidi 2005	57		30	75		30
Recombinant human FSH Study Group 1995	5.0	3.0	53	6.1	3.4	52
Schats 2000	4.3	3.6	220	4.8	4.0	222
Selman 2002	668		133	589		131
Sohrabvand 2012	7.42	3.35	80	7.63	3.28	79
Strowitzki 2016	7	4.18	146	6.5	3.78	153
Westergaard 2001	3.25	0.25	190	3.8	0.4	189
Witz 2020	8.5	5.68	309	5.6	4.31	310
Ye 2012	6.5	3.9	62	4.6	3.0	63

FSH: follicle-stimulating hormone; **rhFSH:** recombinant human follicle-stimulating hormone

Table 3. Number of embryos frozen

Study	Recombinant FSH			Urinary gonadotropins, rhFSH (human cell line), biosimilar		
	Mean	SD	N	Mean	SD	N
Aboulghar 2010	2.57	4.5	42	4.9	6.4	42

Table 3. Number of embryos frozen (Continued)

Andersen 2006	1.9	2.9	368	1.8	2.8	363
Baker 2009	4	4.4	76	2.9	3.3	76
Balasch 2003	1.95	0.65	25	1.52	0.51	25
Bergh 1997	3.2	3.0	119	1.7	2.5	102
Bosch 2008	1.1	1.9	126	1.0	2.1	122
Cheon 2004	484		127	486		121
Frydman 2000	164			68		
Hompes 2008	3.90		209	3.74		198
Hoomans 1999	3.3		83	2.0		82
Lenton 2000	2.3	4.5	80	2.8	4.9	75
Liu 2015	402		243	486		243
Ng 2001	5.75	4.817	20	7.75	5.88	20
Nyboe Andersen 2017	2.2	2.6	643	1.9	2.4	635
Out 1995	2.58			1.81		
Schats 2000	4.3	4.2	220	4.1	4.0	222
Selman 2002	120		133	98		131
Sohrabvand 2012	3.91	2.66	80	4.06	2.56	79
Rashidi 2005	2		30	4		30
Ye 2012	1.9	2.8	62	1.0	1.9	63

FSH: follicle-stimulating hormone; **rhFSH:** recombinant human follicle-stimulating hormone